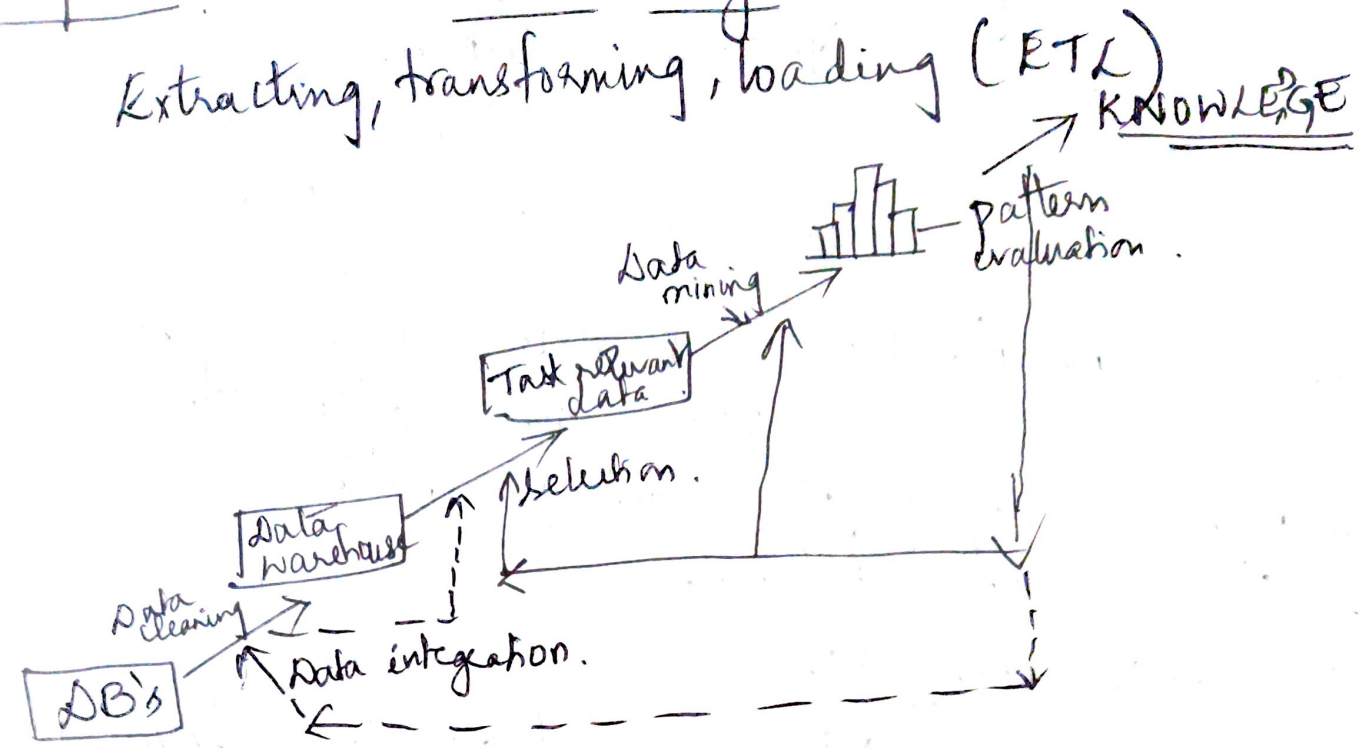
03/06/26

Data Mining.

Extracting, transforming, loading (ETL)



Data Mining in Business Intelligence

— endusa

Decision Making
Data Presentation (Visualization tech) — Business analyst
Data Mining (Info discovery) — Data Analy
Data exploration (Statistical summary, querying)
Data preprocessing/Integration, Data Warehouse — DBA
Data sources

Data preprocessing

— data integration
Normalization
Feature selection.
Dimension reduction

Data Mining

— pattern discovery
Association & correlation
Classification
Clustering
Outlier analysis

Post processing – pattern evaluation.
pattern selection.
pattern interpretation.
pattern visualization

Data to be mined :- DB data, data warehouse, transactional data, stream, text, web, multimedia, graphs, spatiotemporal, time-series, sequence, social & info N/w's.

Techniques utilized :-
Online Analytical processing (data warehouse), data-intensive, ML, statistics, pattern recognition, visualization, high performance etc.

Appln's adapted :- Retail, telecomm°, Banking, fraud analysis, bio-data mining, stock market analysis, text mining, web mining.

Data Mining function :-

1. Generalization
   → info integration & data warehouse construction
   → data cube technology
   → Multidimensional Concept description: characterization & discrimination

2. Association & Correlation Analysis
   → freq patterns
   → Association, correlation vs causality
   → How to mine such patterns & rules efficiently in large data sets ?

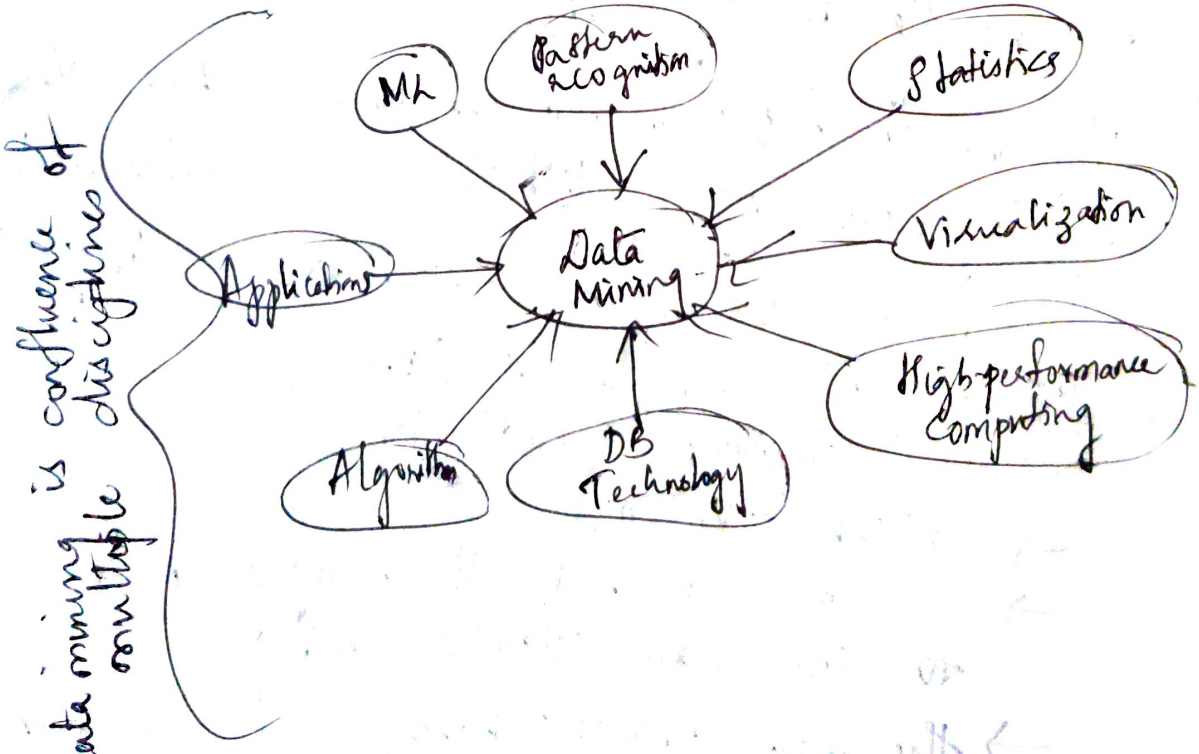
→ How to use such patterns for classification, clustering & other appl^{n}s?

3. Classification
   → Classification & label prediction
   → Typical methods – decision tree, SVM, Neutral N/W etc
   → Typical appl^{n}s – Credit card fraud detection, classifying stars, diseases, web pages

4. Cluster Analysis
   → Unsupervised learning
   → Group data to new categories
   → Principle: Maximizing intra-class similarity & minimizing interclass similarity.

5. Outlier Analysis
   → Methods :- By product of clustering or regression analysis.

Structure & N/W Analysis



Issues in Data Mining

Mining methodology
User Interaction
efficiency & Scalability
Diversity of data types
Data mining & society

Data Preprocessing

Measures for quality data :-

- → Accuracy
- → Completeness
- → Consistency
- → Timeliness
- → Believability
- → Interpretability

Major tasks in Data preprocessing

- → Data cleaning (incomplete, noisy, inconsistent, intentional)
- → Data integration.
- → Data reduction.
- → Data transformation & data discretization (Normalization & concept hierarchy generation)

How to handle missing data?

- → Ignore the tuple - not effective when % of missing values per attribute varies considerably
- → fill missing value manually: tedious + infeasible
- → fill in it automatically with
  - → global constant eg: "unknown"
  - → attribute mean
  - → " " for all samples belonging to the same class: smarter
  - → the most probable value: inference such as Bayesian formula or decision

Noisy data – ?

other data problems which require data cleaning:-
→ duplicate records
→ incomplete data
→ inconsistent data

How to handle Noisy data

1. Binning – first sort data & partition into (equal freq) bins. Then one can smooth by bin means, smooth by bin median, smooth by bin boundaries.
2. Regression – smooth by fitting data into regression functions.
3. Clustering – detect & remove outliers.
4. Combined computer & human inspection.

Data discrepancy detection :-
→ Use metadata
→ Check field overloading
→ " uniqueness rule, consequtive rule, null rule
→ Use Commercial tools :-
→ data scrubbing, data auditing.

Data migration & integration.
→ ETL tools, data migration tools.
→ Integration of 2 processes
→ Iterative & interactive (eg: Potter's wheels)

Data Integration :- 2 approaches :-
→ Schema integration.
→ Instance "

05/06/24 :- Correlation Analysis (Nominal Data) (Numerical data)

X^2 (chi-Square) test for independence :-

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{Expected}}$$

H_0 = Null hypothesis → correlated.
H_a — Alternate hypothesis

If A has c no of values & B has r no of values

then $$\chi^2 = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

$$e_{ij} = \frac{\text{Count}(A = a_i) * \text{Count}(B = b_j)}{n}$$

Degree of freedom = $(c-1)(r-1)$.

① Suppose that group of 1500 ppl were surveyed. Gender of each person was noted. Each person was tolled as to whether each of his/her preffered reading material was fiction or non-fiction.

	M	F	
Fiction	250	200	450
Non-fiction	50	1000	1050
	300	1200	1500

H_0 = Gender & preffered reading are independent

H_1 = Gender & preffered reading are dependent.

$$e_{00} = \frac{\text{Count(male)} * \text{count(fiction)}}{n}$$

$$= \frac{300 * 450}{1500} = 90.$$

$$e_{01} = \frac{\text{Count(female)} * \text{count(fiction)}}{n}$$

$$= \frac{1200 * 450}{1500} = \underline{360}$$

$$e_{10} = \frac{\text{Count (male)} * \text{count (nonfiction)}}{n}$$

$$= \frac{300 * 1050}{1500} = \underline{210}$$

$$e_{11} = \frac{\text{count (female)} * \text{count (nonfiction)}}{n}$$

$$= \frac{1200 * 1050}{1500} = \underline{840}.$$

$$X^2 = \frac{(250-90)^2 + (200-360)^2 + (50-210)^2 + (1000-840)^2}{90 + 360 + 210 + 840}$$

$$= \frac{25600}{90} + \frac{25600}{360} + \frac{25600}{210} + \frac{25600}{840}$$

$$= 284.44 + 71.11 + 121.90 + 30.48$$

$$= 507.93$$

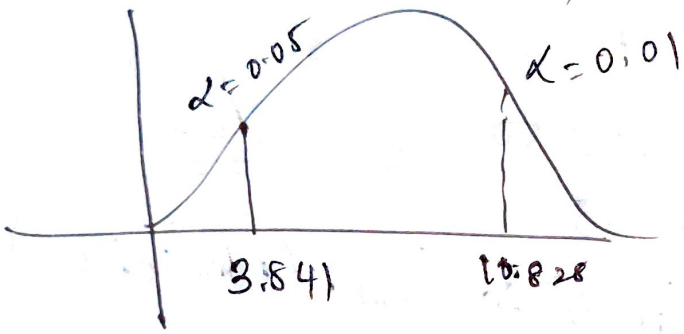
Degree of freedom $= (2-1)(2-1)$

$= 1$

Significance level $\alpha = 0.05$

$= 3.841$

for $\alpha = 0.01 \Rightarrow 10.828$



$507.93 > 10.828$ or 3.84.

Null hypothesis is rejected.

ie, Gender & prefered reading are dependent

Mean :-

$$\text{Mean}(x) = \bar{x} = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$\text{Median}(x) = \begin{cases} x_{(r+1)} & \text{if } m = \text{odd},\ m = 2r+1 \\ \frac{1}{2}(x_r + x_{r+1}) & \text{if } m = \text{even} \\ & m = 2r. \end{cases}$$

$$\text{Variance}(x) = S_x^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2$$

$$\text{Standard Deviation} = \sigma = \sqrt{S_x^2} = \sqrt{\frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2}$$

$\{35, 22, 25, 23, 28, 33, 30\}$

Mean $(\bar{x}) = \bar{X} = \frac{35+22+25+23+28+33+30}{7}$

$= 28$

Median $(\tilde{x})$ = 22, 23, 25, 28, 30, 33, 35

$= 28$

$n = 2r+1$

$7 = 2r+1$

$6/2 = r$

$3 = r$

since n is odd, $x_{(r+1)} = x_{(3+1)}$

$= x_4 = 28$

22, 23, 25, 28, 30, 33, 35 $\quad n = 7$.

Mean

$\bar{X} = 28.$

Variance $S_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

$= \frac{1}{(7-1)} \sum_{i=1}^{7} (22-28)^2 + (23-28)^2 + (25-28)^2 + (28-28)^2 + (30-28)^2 + (33-28)^2 + (35-28)^2$

$= \frac{6^2 + 5^2 + 3^2 + 0 + 2^2 + 5^2 + 7^2}{6}$

S_x or $\sigma^2 = \frac{148}{6} = 24.66$

σ (SD) $= \sqrt{S_x^2} = \sqrt{\text{variance}} = \sqrt{24.66}$

$= \underline{4.96}$

$$\text{Covariance}(x,y) = \frac{1}{n-1}\sum (x_i - \bar{x}) * (y_i - \bar{y})$$

$$\text{Covariance matrix} = \begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(x,y) & \text{var}(y) \end{bmatrix}$$

x_1	x_2
3	7
2	4

$\bar{X}_1 = 5/2 = 2.5$ $\quad \bar{X}_2 = 11/2 = 5.5$

$$\text{var}(X_1) = \frac{1}{m-1}\sum_{i=1}^{m} (X_i - \bar{X}_1)^2$$

$$= \frac{1}{(2-1)}\sum_{i=1}^{2} (3-2.5)^2 + (2-2.5)^2$$

$$0.25 + 0.25 = \underline{0.5}$$

$$\text{var}(X_2) = \frac{1}{1}\sum_{i=1}^{2} (7-5.5)^2 + (4-5.5)^2$$

$$= (1.5)^2 + (1.5)^2$$

$$= 2.25 + 2.25 = \underline{4.5}$$

$$\text{Cov}(x_1, x_2) = \frac{1}{2-1}\left(3-2.5\right) * \left(7-5.5\right) + \left(2-2.5\right) * \left(4-5.5\right)$$

$x_{11} - \bar{x}_1$, $x_{21} - \bar{x}_2$, $x_{12} - \bar{x}_1$, $x_{22} - \bar{x}_2$

$$= (0.5 * 1.5) + (0.5 * 1.5)$$

$$0.75 + 0.75$$

$$= \underline{1.5}$$

Cov Matrix (variance) $= \begin{bmatrix} 0.5 & 1.5 \\ 1.5 & 4.5 \end{bmatrix}$

08/06/26. **Handling Redundancy in Data Integration**

How to identify :—

→ Correlation analysis

→ Covariance analysis

Redundance possible when multiple DB's are integrated.

→ Same attribute may have diff names in diff dB.

→ One attribute may be a "derived" attribute in another table.

Correlation Analysis :- (X^2 test or chi-Square test) :-

Larger X^2 ⇒ more likely, variables are related.

Correlation Co-efficient – also called **pearsons product moment coefficient**.

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A \sigma_B} = \frac{\sum_{i=1}^{n}(a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A \sigma_B}$$

n — no of tuples.

$\bar{A}$ & $\bar{B}$ — means of A, B.

σ_A & σ_B — SD of A & B.

$\Sigma(a_i b_i)$ — sum of AB cross product.

$r_{A,B} = 0$ = independent

$r_{A,B} < 0$ — negatively correlated

$r_{A,B} > 0$ = +vely correlated (A's values increase as B's).

Higher, stronger correlation.

Visually evaluating Correlation:— Using scatter plots

Correlation viewed as linear relationship:
(diff formulas).

Covariance (Numeric Data). It is similar to correlation

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

Correlation Coefficient: $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$

+ve covariance :- If $Cov_{A,B} > 0$

$$Cov(A,B) = E(A \cdot B) - \bar{A}\bar{B}.$$

(2,5), (3,8), (5,10), (4,11), (6,14) ⇒ 2 stocks A & B.

$E(A) = (2+3+5+4+6)/5 = 20/5 = 4$

$E(B) = (5+8+10+11+14)/5 = 48/5 = 9.6$

$E(A \cdot B) = (2*5 + 3*8 + 5*10 + 4*11 + 6*14)$

$= (10+24+50+44+84) = 212$

$\bar{A}\bar{B} = 4 * 9.6 = 38.4$

$E(AB) - \bar{A}\bar{B} \Rightarrow 212 - 38.4 = 173.6$

9.6×4
38.4

Data Reduction Strategies

→ Dimensionality reduction (remove unimportant attributes)

* Wavelet transforms
* PCA
* Feature subset selection, feature creation

→ Numerosity reduction (Data reduction)

* Regression, log-Linear Models.
* Histograms, clustering, sampling,
* Data Cube aggregation.

→ Data compression.

PCA ⇒ finds a projection that captures the largest amount of variation in data. Data is projected to smaller space, leads to dimensionality reduction. We find eigenvectors of covariance matrix. These eigenvectors define new space.

Steps in PCA Algorithm: -

1. Collect Dataset
2. Standardize features
3. Compute covariance matrix
4. Finding eigenvalues & eigenvectors
5. Sort components by explained variance
6. Choose top k components
7. Transform the original dataset

Parametric Data Reduction

→ Linear regression : Data is modeled to fit a straight line.
It uses least-square method to fit the line

→ Multiple regression :

→ Log-linear ~~reg~~ model — approximates discrete multidimensional probability distributions.

Regression Analysis: -

1. Linear regression: - $Y = wX + b$
2. Multiple regression: - $Y = b_0 + b_1 X_1 + b_2 X_2$
3. Log-linear models: - It studies the relationship b/w categorial variables (variable whose values represent groups, categories, labels). using the counts (freq) in contingency table.

eg: - Gender = {male, female}
Blood Grp — O+ve, A+ve, AB+, etc---
Dept - CS, IS, EC etc.

Unit-2 Def^n:-

Entropy, Gini Index, InfoGain, Tree pruning with ex

Entropy :- measure of impurity or uncertainty in a dataset. It is used in ML for building decision trees.

Entropy tells how mixed classes are in the dataset

Low entropy (0) - all records belong to one class (pure dataset)

High entropy (1 for two classes) - classes are equally mixed. (max uncertainty)

$$\text{Entropy}(S) = -\sum_{i=1}^{n} P_i \log_2(P_i)$$ where S - dataset

P_i - probability of class i.

ex1: pure dataset.

Suppose a dataset has 10 Yes & 0 No.

$P(\text{Yes}) = 1$, $P(\text{No}) = 0$

Entropy $= -(1 * \log_2 1 + 0 * \log_2 0) = 0 \Rightarrow$ dataset is completely pure

ex 2: Mixed dataset. Dataset has 5 Yes & 5 No

entropy $= -(0.5 \log_2 0.5 + 0.5 \log_2 0.5)$

$= \log_2(0.5) = -1$

entropy $= -(0.5 * -1 + 0.5 * -1) = 1 \Rightarrow$ meaning dataset is highly impure.

Info gain :- measure used in decision tree algorithms to select best attribute for splitting dataset.

$$IG(S, A) = \text{Entropy}(S) - \sum_{v \in A} \frac{|S_v|}{|S|} * \text{Entropy}(S_v)$$

where S - original dataset

A - attribute for splitting

S_v - Subset for each value of attribute A

$|S_v| / |S|$ = Weight of each subset.

ex :- 10 records - 5-Yes, 5-No.

(written in diff book & the rest)

→ Histogram Analysis – divide data into buckets & store avg.

→ Clustering – Partition data set into clusters based on similarities

→ Sampling – obtaining small sample s to represent whole data set N.

Types:

* Simple random sampling
* Sampling without replacement
* Sampling with replacement
* Stratified sampling.

Data transformation:– maps entire set of values of a given attribute to a new set of replacement values. so that each old value can be identified with one of the new values.

Methods:- * Smoothing – remove noise from data.
* Attribute/feature construction.
* Aggregation
* Normalization.

Normalization:-

1. Min-max normalization : to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

eg: income range $12000 to $98,000.
→ normalized to [0.0, 1.0]

If $v = \$73,600$

Then $= \frac{73,600 - 12000}{98,000 - 12000}(1.0 - 0) + 0 = 0.716$

(2) Z-score normalization :- (μ : Mean, σ : SD):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

If $\mu = 54000$, $\sigma = 16,000$

Then $\frac{73,600 - 54000}{16000} = 1.225$

(3) Normalization by decimal scaling: -

$v' = \frac{v}{10^j} \Rightarrow$ j is the smallest integer such that $Max(|v'|) < 1$

Discretization: - divide the range of continuous attribute into intervals.

3 types of attributes :-

1. Nominal — values from unordered set eg: color, profession
2. Ordinal — values from ordered set eg: military rank
3. Numeric — real no

Methods :- 1) Binning < equal-width (distance) partitioning
equal-depth (frequency) partitioning

partition into equal freq:-

Smoothing by bin means

bin 1 : 9, 9, 9, 9
bin 2 : 23, 23, 23, 23
bin 3 : 29, 29, 29, 29

Smoothing by bin boundaries

bin 1 : 4, 4, 4, 15
bin 2 : 21, 21, 25, 25
bin 3 : 26, 26, 26, 34

eg:- 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

bin 1 : 4, 8, 9, 15
bin 2 : 21, 21, 24, 25
bin 3 : 26, 28, 29, 34

09/6/26

Unit-2

Data Mining — Frequent pattern analysis

Frequent pattern — A pattern that occurs frequently in a dataset.

↳ used to find inherent regularities in data.

Appl^s :- Basket data analysis, cross-marketing, DNA sequence analysis, catalog design

pattern can be a set of items, subsequences, substructures etc

Frequent pattern analysis was proposed in the context of **frequent itemsets** & **association rule mining**

Frequent pattern mining is important bcoz frequent pattern is an intrinsic & important property of datasets.

→ It is the foundation for many essential data mining tasks like Association, correlation, causality analysis etc etc etc - - -.

Tid	Items Bought
10	Beer, Nuts, Diaper.
20	Beer, Coffee, Diaper
30	Beer, Diaper, eggs
40	Nuts, eggs, milk
50	Nuts, Coffee, Diaper, Egg, Milk.

Freq patter :- Beer: 3, Nuts: 3, Diapes: 4, Eggs: 3, {Beer, Diaper}: 3

Association rules: — (support, confidence)

Beer → Diaper (60%, 100%)

Diaper → Beer (60%, 75%)

Support (s) — probability that a transactions contains X ∪ Y

Confidence (c) conditional probability that a transaction X also has Y.

Confidence (A ⇒ B) = Support (A ∪ B) / Support (A)

Closed Patterns & Max patterns

Whenever there are too many patterns or long pattern — then there is need for closed patterns & max patterns

TID	Items
1	A, B, C
2	A, B
3	A, C
4	B, C

Support (A) = 3 {because it comes 3 times

" (A, B) = 2

" (A, C) = 2

" (B, C) = 2

" (A, B, C) = 1

" (B) = 3

" (C) = 3

Since support ({A,B,C}) = 1 < 2, it is not frequent
Assume, min support = 2

A → support = 3.

Supersets {A,B} & {A,C} have support 2, not 3.
so A is closed.
Illarly for B & C.

{A,B} → support = 2.
Superset {A,B,C} has support 1.
So AB, AC, BC are all closed.

Rule: Item set is closed if there is no superset having the same support.

Max freq patterns :- A frequent itemset is max if none of its supersets are frequent.

{A}, {B}, {C} — not maximal bcoz they have frequent supersets.

{A,B} — has no freq superset.
{A,C} — " " " "
{B,C} — " " " "
Max patterns :- {A,B}, {A,C}, {B,C}.

Z-Score :

Mean Absolute Deviation = S_A

$$S_A = \frac{1}{n}\left(|V_1 - \bar{A}|\right) + |V_2 - \bar{A}| + |V_3 - \bar{A}| + \ldots + |V_n - \bar{A}|\right)$$

$$V_i' = \frac{V_i - \bar{A}}{S_A}$$

Normalization using decimal scaling.

$$V_i' = \frac{V_i}{10^j} \text{ where } \max(|V_i'|) \leq 1$$

ex:- If A = −986 to 917,

then $\frac{986}{1000} = 0.98 < 1$. $\therefore j = 3$

since $10^3 = 1000$

Given data :— V_1 V_2 V_3 V_4 V_5

200, 300, 400, 600, 1000.

a. MinMax Normalization by setting min = 0, max = 1

b. Z score Normalization

c. Z score Normalization using Mean Absolute Value

d. Normalization using decimal scaling.

a. $$V_i' = \frac{V_i - \min_A}{\max_A - \min_A}\left(new_max_A - new_min_A\right) + new_min_A$$

$new_min_A = 0$, $new_max_A = 1$

$$200' = \frac{200 - 200}{1000 - 200}\cdot(1 - 0) + 0 = 0$$

$$300^1 = \frac{300-200}{1000-200}(1-0)+0 = \frac{100}{800} = \frac{1}{8}$$

$$= 0.125$$

$$400^1 = \frac{400-200}{1000-200}(1-0)+0 = \frac{200}{800} = \frac{1}{4} = 0.25$$

$$600^1 = \frac{600-200}{1000-200}(1-0)+0 = \frac{400}{800} = \frac{1}{2} = 0.5$$

$$1000^1 = \frac{1000-200}{1000-200}(1-0)+0 = \frac{800}{800} = 1$$

So the new set :- $A^1 = \{0, 0.125, 0.25, 0.5, 1\}$

These values are normalized now.

2. Z-score normalization :-

$$v_i^1 = \frac{v_i - \mu_A}{\sigma_A} \text{ or } \bar{A} \quad \text{where } \mu = \text{mean} \ \& \ \sigma = \text{SD}$$

Mean = $200+300+400+600+1000 / 5 = 500$

$$\sigma_A = \sqrt{\frac{\Sigma (x_i - \mu)^2}{n}}$$

$(x+\bar{A})^2$ or μ^2
$(200-500)^2 = 300^2$
$(300-500)^2 = 200^2$
$(400-500)^2 = 100^2$
$(600-500)^2 = 100^2$
$(1000-500)^2 = 500^2$

$$\sigma_A = \sqrt{\frac{1}{5}(300^2+200^2+100^2+100^2+500^2}$$

$$= \sqrt{80000} = 282.84$$

v_1' for 200 :- $\frac{200-500}{282.84} = -1.06$

v_2' for 300 :- $\frac{300-500}{282.84} = -0.707$

v_3' for 400 :- $\frac{400-500}{282.84} = -0.35$

v_4' for 600 :- $\frac{600-500}{282.84} = 0.354$

v_5' for 1000 :- $\frac{1000-500}{282.84} = 1.77$

$v_i = \{200, 300, 400, 600, 1000\}$

$v_i' = \{-1.06, -0.707, -0.35, +0.354, +1.77\}$

3. Z-Score Normalization using Mean Absolute Value

$$v_i' = \frac{v_i - \bar{A}}{S_A}$$

$$S_A = \frac{1}{n}\left(|v_1 - \bar{A}| + |v_2 - \bar{A}| + |v_3 - \bar{A}| + \cdots + |v_n - \bar{A}|\right)$$

$$= \frac{1}{5}\left(|200-500| + |300-500| + |400-500| + |600-500| + |1000-500|\right)$$

$$= \frac{1}{5}(1200) = 240$$

$$v_1' = \frac{200-500}{240} = -1.25$$

$$v_2' = \frac{300-500}{240} = -0.833$$

$$v_3' = \frac{400-500}{240} = -0.4166$$

$$v_4' = \frac{600-500}{240} = 0.417$$

$$v_5' = \frac{1000-500}{240} = 2.08$$

$v_i = \{200, 300, 400, 600, 1000\}$

$v_i' = \{-1.25, -0.833, -0.417, +0.417, +2.08\}$

d. Normalization by decimal scaling :-

$$v_i' = \frac{v_i}{10^j} \quad \text{where } \max |v_i'| \le 1$$

$j = 3$

because $10^3 = 1000$

$$v_1' = \frac{200}{10^3} = 0.2$$

$$v_2' = \frac{300}{10^3} = 0.3$$

$$v_3' = \frac{400}{10^3} = 0.4$$

$$v_4' = \frac{600}{10^3} = 0.6$$

$$v_5' = \frac{1000}{10^3} = 1.0$$

$v_i' = \{0.2, 0.3, 0.4, 0.6, 1.0\}$

Discretization

4, 8, 15, 21, 21, 24, 25, 28, 34 → Freq = 3

1st sort, then apply if not sorted

~~Partition into equisized bin~~

a. Equal frequency.
b. Smoothing by bin mean.
c. Smoothing by bin boundary.

a. equal freq:-

Bin 1 – 4, 8, 15
Bin 2 – 21, 21, 24
Bin 3 – 25, 28, 34

b. Smoothing by bin mean:-

~~Mean = 4+8+15+21+21+24+25+28+34/9~~

Bin 1 mean = 4+8+15/3 = 27/3 = 9

Bin 2 mean = 21+21+24/3 = 22

Bin 3 mean = 25+28+34/3 = 87/3 = 29.

(4) (8) (15)

Bin 1: 9, 9, 9.
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

c. Smoothing by bin boundary

Bin 1 – 4, 4, 15
Bin 2 – 21, 21, 24
Bin 3 – 25, 25, 34

4 is near to 4
8 is near to 4
15 is near to 15
Illarly for others.

(2) 5, 10, 11, 13, 15, 35, 50, 56, 72, 92, 204, 215

Total = 12, Frequency = 4.

So 3 bins.

a. Equal freq:-

bin 1 :- 5, 10, 11, 13

bin 2 :- 15, 35, 50, 56

bin 3 : 72, 92, 204, 215

b. Smoothing by bin mean.

bin 1 :- 39/4 = 9.75

bin 2 = 156/4 = 114

bin 3 :- 583/4 = 145.75

bin 1 : 9.75, 9.75, 9.75

bin 2 : 114, 114, 114

bin 3 : 145.75, 145.75, 145.75

c. Smoothing by bin boundary

bin 1 : 5, 13, 13, 13

bin 2 : 15, 15, 56, 56.

bin 3 : 72, 72, 215, 215

10/06/26 Apriori Algorithm — for pattern mining

↓

It gives frequent pattern

Dataset D

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

$Sup_{min} = 2$ {min support = 2}

Scan Dataset (D) for count of each candidate

ItemSet	Sup. count	Compare candidate support count with min support count
		satisfied ✓
A	2	✓
B	3	✓
C	3	✓
D	1	X
E	3	✓

L_1 Itemset	Support count.
A	2
B	3
C	3
E	3

⇒ Generate C_2 Candidate. from L_1.
ie $L_1 * L_1$

C_2 ItemSet :-

	Support count
A B	1
A C	2
A E	1
B C	2
B E	3
C E	2

✓ all satisfies min support

[Coz, AB appears together only once, AC " " twice & so on.]

Now compare candidate support with min support count.

ie AB - X, AC ✓, AE X, BC - ✓, BE - ✓, CE - ✓

So satisfied :- AC, BC, BE, CE ⇒ these are items selected for next level

Items	support Count
AC	2
BC	2
BE	3
CE	2

C_3 Itemset ⇒ from $L_2 * L_2$

support count

ABC = 1 X
ACE = 1 X
BCE = 2 ✓ only this satisfies
ABE = 1 X

ABC, ABCE, ACE, BCE, ABE

ABC ✓ already sup.
BCE ✓ "
ACE ✓ "
ABE

When Compared with min support count.
only BCE — satisfies condn.

Association Rules :-

there are only that satisfies support count.

$A \to C$, $C \to A$, $B \to C$, $C \to B$, $B \to E$, $E \to B$, $BC \to E$, $CE \to B$, $BE \to C$

for $A \to C$, $C \to A$

Confidence $(A \to C) = \frac{\text{Support count } (A \cup C)}{\text{Support count } (A)}$

$= \frac{2}{2} = 100\%$.

Confid $(C \to A) = \frac{\text{Sup count } (C \cup A)}{\text{Sup count } (C)}$

$= 2/3 = 0.66$

Since its less than (90%) — threshold, we skip this.

for $B \to C$, $C \to B$

$$\text{Confidence}(B \cup C) = \frac{\text{Supcount}(B \cup C)}{\text{Supcount}(B)} = \frac{2}{3} = 0.66$$

$$\text{Confidence}(C \cup B) = \frac{\text{Supcount}(B \cup C)}{\text{Supcount}(C)} = \frac{2}{3} = 0.66$$

for $BC \to E$ &

$$\text{Confidence}((BC) \cup E) = \frac{\text{Supcount}((BC) \cup E)}{\text{Supportcount}(BC)} = \frac{2}{2} = 1$$

~~Confidence~~ $(E \cup (BC)) = \frac{\text{Supcount}(E \cup (BC))}{\text{Supportcount}(E)} = \frac{2}{3} = 0.66$

$BE \to C$

$$\text{Confid}((BE) \cup C) = \frac{\text{Supcount}((BE) \cup C)}{\text{Supcount}(BE)} = \frac{2}{3} = 0.66$$

We must consider high confidence value. (usually more than 90%)

[If subsets are not supporting min support count — discard it. This is called pruning].

(2)

Tid	List of items
T_{100}	I_1, I_2, I_5
T_{200}	I_2, I_4
T_{300}	I_2, I_3
T_{400}	I_1, I_2, I_4
T_{500}	I_1, I_3
T_{600}	I_2, I_3
T_{700}	I_1, I_3
T_{800}	I_1, I_2, I_3, I_5
T_{900}	I_1, I_2, I_3

a. Find the freq patterns min sup = 2.

b. Find confidence value & 2 itemsets & 3 itemsets.

Confidence value threshold = 90%.

L_1

Items	Support count	
I_1	6	✓
I_2	7	✓
I_3	6	✓
I_4	2	✓
I_5	2	✓

All satisfy min support

Now $L_1 * L_1$

Items	Support Count	
I_1, I_2	4	
I_1, I_3	4	
I_1, I_4	1	X
I_1, I_5	2	
I_2, I_3	4	
I_2, I_4	2	
I_2, I_5	2	
I_3, I_4	0	X

I_3, I_5	1	X
I_4, I_5	0	X

Items that satisfy is

L_2

I_1, I_2	4
I_1, I_3	4
I_1, I_5	2
I_2, I_3	4
I_2, I_4	2
I_2, I_5	2

$I_1, I_2,$
I_3, I_4

$I_1, I_3,$
I_2, I_5

$\{I_1, I_3\}, \{I_2, I_4\}$
$\{I_1, I_5\}, \{I_2, I_3\}$
$\{I_1, I_5\}, \{I_2, I_4\}$
$\{I_1, I_5\}, \{I_2, I_5\}$
I_1, I_2, I_5

Now $L_2 * L_2$

	Support count
I_1, I_2, I_3	2 ✓
I_1, I_2, I_5	2 ✓
I_1, I_2, I_4	1
I_1, I_3, I_5	1
I_1, I_3, I_4	0
I_2, I_3, I_4	0
I_2, I_3, I_5	1
I_2, I_4, I_5	0

So items ~~cnt~~ now are :- } both supports min supp count.

L_3

I_1, I_2, I_3	2
I_1, I_2, I_5	2

L_4 :-

I_1, I_2, I_3, I_5	1	— X doesn't support

So final values/rules:-

$\{I_1, I_2\} \Rightarrow I_1 \to I_2, \quad I_2 \to I_1$

$\{I_1, I_3\} \Rightarrow I_1 \to I_3, \quad I_3 \to I_1$

$\{I_1, I_5\} \Rightarrow I_1 \to I_5, \quad I_5 \to I_1$

$\{I_2, I_5\} \Rightarrow I_2 \to I_5, \quad I_5 \to I_2$

$\{I_2, I_3\} \Rightarrow I_2 \to I_3, \quad I_3 \to I_2$

$\{I_2, I_4\} \Rightarrow I_2 \to I_4, \quad I_4 \to I_2$

$\{I_1, I_2, I_3\} \Rightarrow I_1, I_2 \to I_3, \quad I_2, I_3 \to I_1, \quad I_1, I_3 \to I_2$

$\{I_1, I_2, I_5\} \Rightarrow I_1, I_2 \to I_5, \quad I_1, I_5 \to I_2, \quad I_2, I_5 \to I_1$

$$\text{Conf}(I_1 \cup I_2) = \frac{\text{Supcount}(I_1 \cup I_2)}{\text{Supcount}(I_1)}$$

$$= \frac{4}{6} = 0.66$$

$$\text{Conf}(I_2 \cup I_1) = \frac{4}{7} = 0.57$$

$$\text{Conf}(I_1 \cup I_3) = \frac{4}{6} = 0.66$$

$$\text{Conf}(I_3 \cup I_1) = \frac{4}{6} = 0.66$$

$$\text{Conf}(I_1 \cup I_5) = \frac{2}{6} = 0.33$$

$$\text{Conf}(I_5 \cup I_1) = \frac{2}{2} = 1 \quad \checkmark$$

$$\text{Conf}(I_2 \cup I_5) = \frac{2}{7} = 0.28$$

$$\text{Conf}(I_5 \cup I_2) = \frac{2}{2} = 1 \quad \checkmark$$

$\text{Conf}(I_2 \cup I_3) = \frac{4}{4} = 0.57$ ~~2~~

$\text{Conf}(I_3 \cup I_2) = \frac{4}{6} = 0.66$

$\text{Conf}(I_2 \cup I_4) = \frac{2}{4} = 0.28$

$\text{Conf}(I_4 \cup I_2) = \frac{2}{2} = 1$ ✓

$\text{Conf}((I_1, I_2) \cup I_3) = \frac{2}{4} = 0.5$

$\text{Conf}((I_2, I_3) \cup I_1) = \frac{2}{4} = 0.5$

$\text{Conf}((I_1, I_3), \cup I_2) = \frac{2}{4} = 0.5$

$\text{Conf}((I_1, I_2) \cup I_5) = \frac{2}{4} = 0.5$

$\text{Conf}((I_1, I_5), \cup I_2) = \frac{2}{2} = 1$ ✓

$\text{Conf}((I_2, I_5) \cup I_1) = \frac{2}{2} = 1$ ✓

11/6/26

FP - Growth Algorithm

To overcome the drawbacks of Apriori algorithm this algorithm is introduced.

Tid	List of the items
T_{100}	I_1, I_2, I_5
T_{200}	I_2, I_4
T_{300}	I_2, I_3
T_{400}	I_1, I_2, I_4
T_{500}	I_1, I_3
T_{600}	I_2, I_3
T_{100}	I_1, I_3
T_{800}	I_1, I_2, I_3, I_5
T_{900}	I_1, I_2, I_3

Min Support = 2.

Step1:- Find C_1 candidate and its support count

Item	Supp Count
I_1	6
I_2	7
I_3	6
I_4	2
I_5	2

Step 2 :- Set the language L.
(Check the item with higher freq, arrange in desc order)

$L : \{ I_2 : 7, I_1 : 6, I_3 : 6, I_4 : 2, I_5 : 2 \}$.

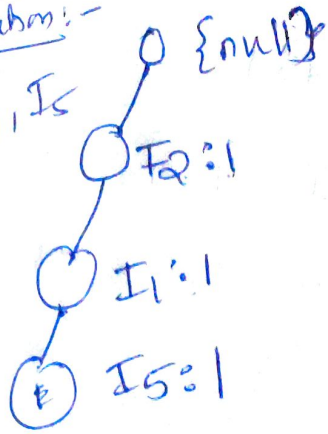
Step 3: Now sort the ele in table according to descending order.

T_{100}	I_2, I_1, I_5
T_{200}	I_2, I_4
T_{300}	I_2, I_3
T_{400}	I_2, I_1, I_4
T_{500}	I_1, I_3
T_{600}	I_2, I_3
T_{700}	I_1, I_3
T_{800}	I_2, I_1, I_3, I_5
T_{900}	I_2, I_1, I_3

Step 4:- Tree Construction.

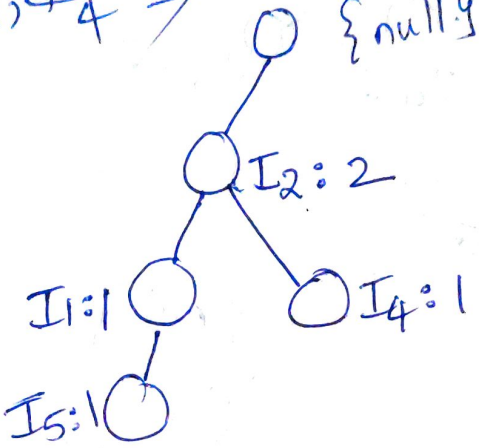
1st transaction:-

I2, I1, I5

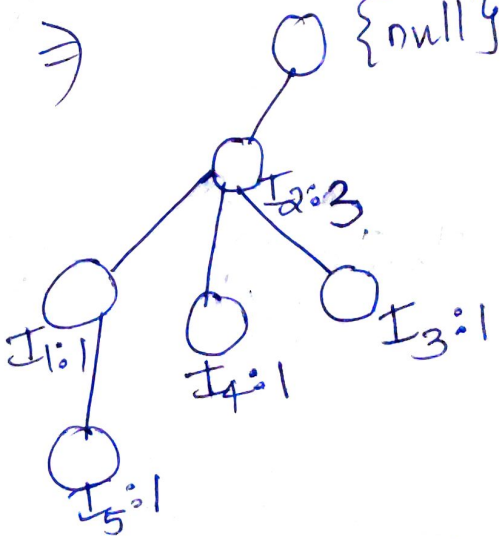


I2:1
I1:1
I5:1
} bcoz we have visited them only once.

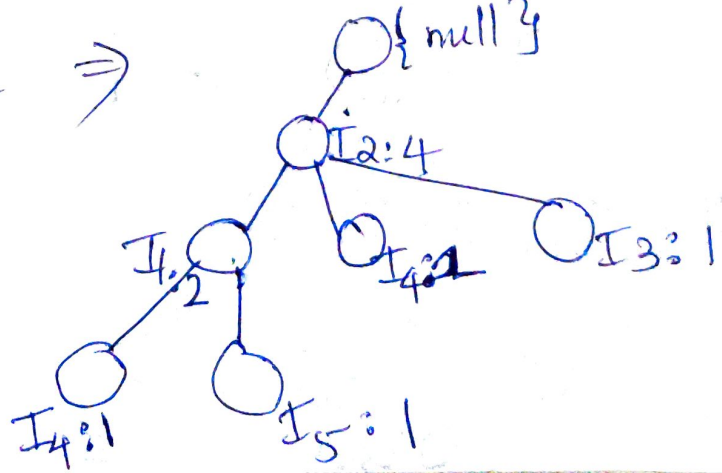
2nd :- I2, I4 ⇒



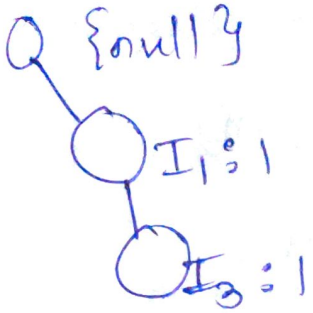
3rd: I2, I3 ⇒



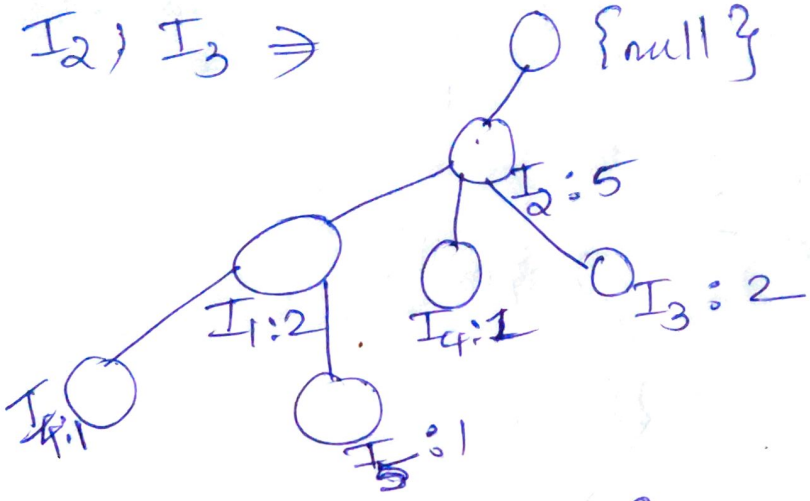
4th. I2, I1, I4 ⇒



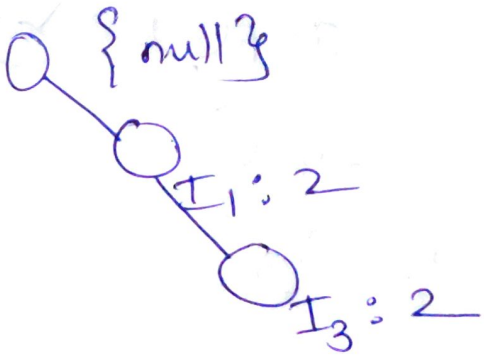
5th : $I_1, I_3 \Rightarrow$



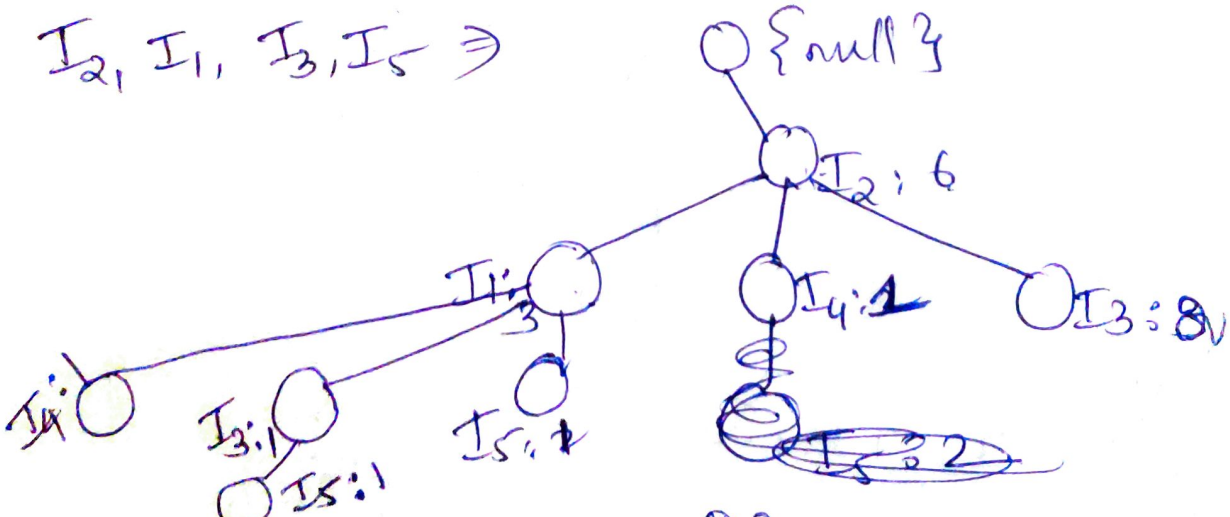
6th :- $I_2, I_3 \Rightarrow$



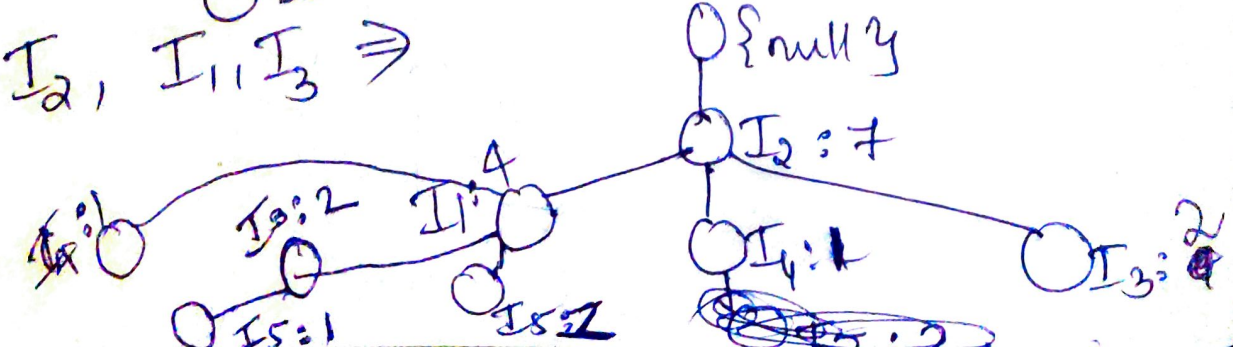
7th : $I_1, I_3 \Rightarrow$



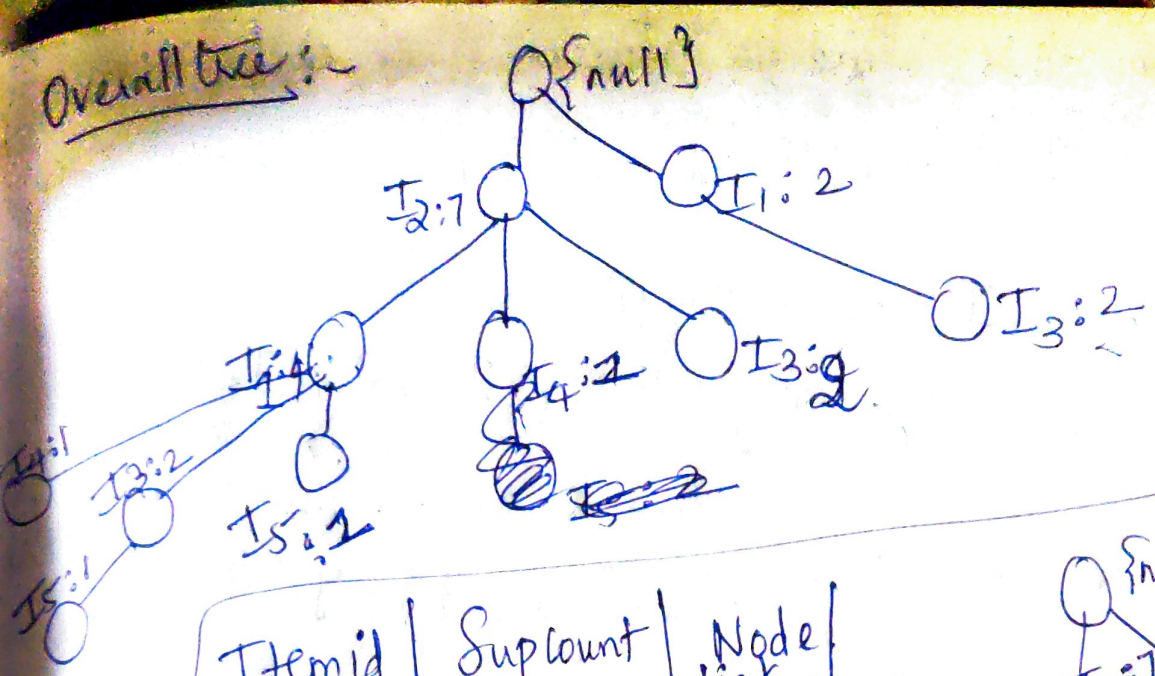
8th : $I_2, I_1, I_3, I_5 \Rightarrow$



9th : $I_2, I_1, I_3 \Rightarrow$

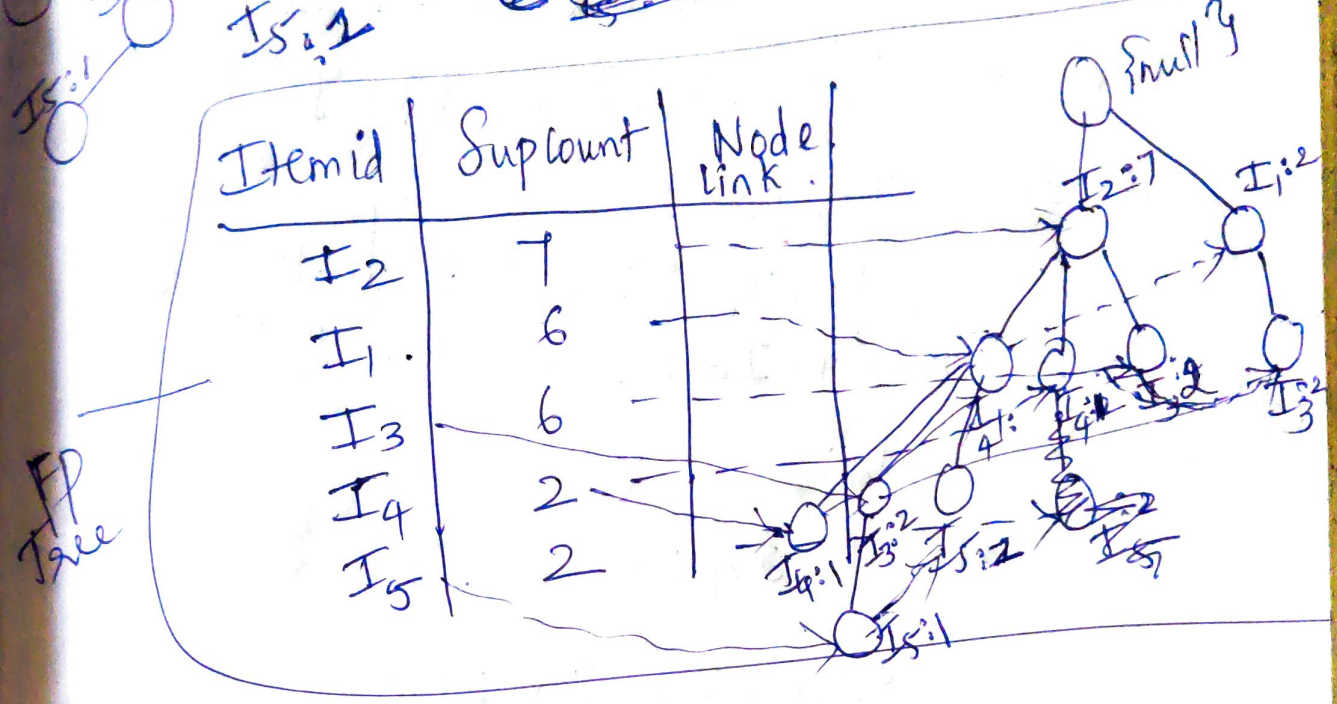


Overall tree :-



Item id	Sup count	Node link
I_2	7	
I_1	6	
I_3	6	
I_4	2	
I_5	2	

FP Tree



FP Mining :-

1. Start from each freq length = 1 pattern.
Construct Conditional pattern base.

Last item :- I_5 :

$\langle I_2, I_1, I_5 : 1 \rangle$

$\langle I_2, I_1, I_3, I_5 : 1 \rangle$

Suffix of I_5 =

$\langle I_2, I_1 \rangle$

$\langle I_2, I_1, I_3 \rangle$

How we reach I_5

$\{I_2, I_1, I_5\} : 1$

$\{I_2, I_1, I_3, I_5\} : 1$

$\{I_2, I_1\}$

$\{I_2, I_1, I_3\}$

Conditional pattern base of I_5

From this find no. of occurances of each item.

$I_2 : 2$, $I_1 : 2$, $I_3 : 1$

$I_3 <$ min supp.

So we only take $I_2 : 2$, $I_1 : 2$

This is called conditional FP Tree

Now 2 sets :- $\{I_2, I_1\}$, $\{I_5\}$

So ⇒ $I_2, I_5 : 2$
$I_5, I_1, I_2 : 2$
$I_1, I_5 : 2$

} frequent pattern generation.

Next

Last Item : I_4 :-

Path of I_4 :-

$\langle I_2, I_1, I_4 : 1 \rangle$

$\langle I_2, I_4 : 1 \rangle$

Suffix of I_4 :- $\{I_2, I_1\}$

$\{I_2\}$

$I_2 : 2$, $I_1 : 1$ — ~~Conditional FP tree~~

Since $I_1 <$ minsupp, we take only $\langle I_2 : 2 \rangle$ → Condnal FP tree

Now sets :- $\{I_2\}$ ~~$\{I_4\}$~~

So we get ⇒ ~~$\{I_2, I_4\}$~~ $\{I_2, I_4\} : 2$ → freq pattern

Next item :- I_3 :-

path of I_3 :- $\langle I_2, I_1, I_3 : 2 \rangle$

$\langle I_2, I_3 : 2 \rangle$

$\langle I_1, I_3 : 2 \rangle$

Suffix of I_3 :- $\{I_2, I_1\}$

$\{I_2\}$

$\{I_1\}$.

$I_2 : 2, I_1 : 2$

~~Since I_1 < min support skip it~~.

$I_1 : 2, I_2 : 2 \Rightarrow$ Condⁿal FP tree.

Now sets :- ~~$\{I_1\}, \{I_2\}, \{I_3\}$~~

$\{I_1, I_2\}, \{I_3\} \Rightarrow \{I_2, I_3\} : 2$ ⇒ freq pattern generation

$\{I_1, I_3\} : 2$

$\{I_1, I_2, I_3\} : 2$

Next item :- I_1 : path :- $\langle I_2, I_1 : 4 \rangle$

$\langle I_1 : 2 \rangle$

~~$I_2, I_1 : 4$~~

Suffix of I_1 : $I_2 : 4$

Next path.

I_2 : Path : $\{I_2 : 7\}$

$$\text{Conf}(A \cup B) = \frac{\text{Supcount}(A \cup B)}{\text{sp.Count}(A)}$$

$$\text{Sup}(A) = \frac{\text{freq}(A)}{\text{no of transactions}}$$

An itemset is closed if its superset has a different frequency than its item set.

A maximal frequent itemset – is a frequent itemset that has no freq superset.

eg:- Assume min sup = 2

A = 3, B = 3, C = 3 — not max itemset bcoz if has freq superset AB, AC, BC.

AB = 2, AC = 2, BC = 2.

ABC = 1 — (not freq)

So Max itemsets = {AB, AC, BC}

since no frequent superset

12/6/26

Mining the frequent pattern using vertical format:- (It is a variation of Apriori)

Tid	item-id
T_{100}	I_1, I_2, I_5
T_{200}	I_2, I_4
T_{300}	I_2, I_3
T_{400}	I_1, I_2, I_4
T_{500}	I_1, I_3
T_{600}	I_2, I_3
T_{700}	I_1, I_3
T_{800}	I_1, I_2, I_3, I_5
T_{900}	I_1, I_2, I_3

min support = 2

Step 1:-

L_1 (Itemset)	TID-set
✓ $\{I_1\}$	$T_{100}, T_{400}, T_{500}, T_{700}, T_{800}, T_{900}$
✓ $\{I_2\}$	$T_{100}, T_{200}, T_{300}, T_{400}, T_{600}, T_{800}, T_{900}$
✓ $\{I_3\}$	$T_{300}, T_{500}, T_{600}, T_{700}, T_{800}, T_{900}$
✓ $\{I_4\}$	T_{200}, T_{400}
✓ $\{I_5\}$	T_{100}, T_{800}

All 5 satisfies Condn. minsupp = 2.

L_2.

Itemset	TID's.
✓ $\{I_1, I_2\}$	$T_{100}, T_{400}, T_{800}, T_{900}$
✓ $\{I_1, I_3\}$	$T_{500}, T_{700}, T_{800}, T_{900}$
✗ $\{I_1, I_4\}$	T_{400}
✓ $\{I_2, I_3\}$	$T_{300}, T_{600}, T_{800}, T_{900}$
✓ $\{I_2, I_4\}$	T_{200}, T_{400}
✓ $\{I_2, I_5\}$	T_{100}, T_{800}
$\{I_3, I_4\}$	—
$\{I_3, I_5\}$	T_{800}
$\{I_4, I_5\}$	—

only these satisfy the Condn. minsupp=2

L_2. So take only $\{I_1, I_2\}, \{I_1, I_3\}, \{I_2, I_3\}, \{I_2, I_4\}, \{I_2, I_5\}$

L_3:

Itemset	TID's.
✓ I_1, I_2, I_3	T_{800}, T_{900}
I_1, I_2, I_4	T_{400}
✓ I_1, I_2, I_5	T_{100}, T_{800}
I_1, I_3, I_4	∅
I_1, I_3, I_5	T_{800}
I_2, I_3, I_4	∅

I_2, I_3, I_5	800
I_2, I_4, I_5	0

Only consider $\{I_1, I_2, I_3\}$ & $\{I_1, I_2, I_5\}$

4 item set :- $\{I_1, I_2, I_3, I_5\}$ = 800

But it doesn't satisfy min support.

Final rules :-

$I_1 \rightarrow I_2$, $I_2 \rightarrow I_1$

$I_1 \rightarrow I_3$, $I_3 \rightarrow I_1$

$I_1 \rightarrow I_5$, $I_5 \rightarrow I_1$

$I_2 \rightarrow I_3$, $I_3 \rightarrow I_2$

$I_2 \rightarrow I_4$, $I_4 \rightarrow I_2$

$I_1, I_2 \rightarrow I_3$, $I_2, I_3 \rightarrow I_1$, $I_1, I_3 \rightarrow I_2$

$I_1, I_2 \rightarrow I_5$, $I_1, I_5 \rightarrow I_2$, $I_2 I_5 \rightarrow I_1$

Q.

TID	items bought
100	{f, a, c, d, g, i, m, p}
200	{a, b, c, f, l, m, o}
300	{b, f, h, j, o, w}
400	{b, c, k, s, p}
500	{a, f, c, e, l, p, m, n}

→ Construct FP-tree

→ Find out frequent patterns ~~association~~

→ Find the support count confidence of each association rule derived from the FP Growth Algorithm.

1. Find all candidates & find its supportcount;

Items	Supp count.
✓ a	3
✓ b	3
✓ c	4
d	1
e	1
✓ f	4
g	1
h	1
i	1
j	1
K	1
l	2
✓ m	3
n	1
o	2
✓ p	3
~~w~~	~~1~~

min Supcount = 3

w	1
s	1

Only a, b, c, f, m, p. satisfy cond".

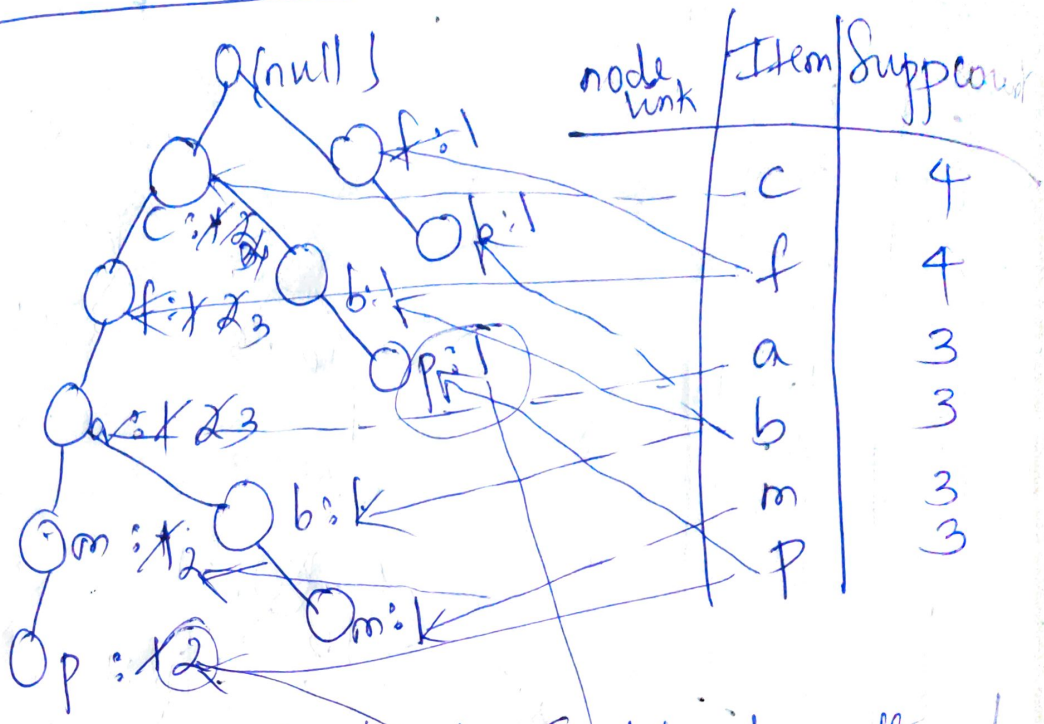
Now arrange the item in decreasing order of frequency

L : { c : 4, f : 4, a : 3, b : 3, m : 3, p : 3 }

Construct the table according to higher freq.

TID	items brought
100	~~f~~, c, f, a, m, p, ~~d, g, i~~
200	~~f~~, c, f, a, b, m, ~~k, o~~
300	f, b, ~~o, h, j, w~~
400	c, b, p, ~~k, s~~
500	f, c, a, p, m, ~~f, e, o~~

Tree construction :-



FP mining :- Construct Conditional pattern base

Last item :-

Condnal pattern base of P

P : 3 = {c, f, a, m, p : 2}

{c, b, p : 1}

Suffix of P :- {c, f, a, m : 2 ; c, b : 1} ⇒ c = 2, c = 1

c : 3, f : 2, a : 2, m : 2, ∴ c = 2 + 1 = 3

b : 1

Only c supports minsupport.

∴ sets are :- {c, P : 3}

Condnal pattern base of m : 3

{c, f, a, m : 2}

{c f, a, b, m : 1}

Suffix of m :- <c, f, a : 2>

<c, f, a, b : 1>

∴ c : 3, f : 3, a : 3, b : 1

b doesn't support

So sets are :-

{c}, {f}, {a}, {m}

freq items :- <caf, m : 3>
<f, m : 3> <c, m : 3> <ca, m : 3>
<f, m : 3> <af, m : 3> <a, m : 3>

⇒ {c, f}, {c, a}, {c, m}, {a, f}, {m, f}
{a, m}, {c, a, f}, {c, m, f}, {f, a, m}
{c, a, m}

So freq <c, m : 3>, <f, m : 3>, <a, m : 3>, <c, m, f : 3>,
<c, a, m : 3>, <a, f, m : 3>, <c, a, f, m : 3>

Condnal pattern base of b :-

{c, f, a : 1}

{f : 1}

{c : 1}

Suffix of b :-

c : 2, f : 2, a : 1

No pattern will be generated

Conditional pattern base of a :-

{c, f, a : 3}

Suffix of a

c : 3, f : 3, cf : 3

Combinations are {c, a : 3}, {f, a : 3}, {c, f, a : 3}

Cond^nal pattern base of f :-

{c, f : 3}, {f : 1}

Suffix of f : c = 3

∴ combination = {f, c : 3}

Items	Conditional pattern Base	Cond^nal FP	Freq pattern gen
{p}	{c, f, a, m, p : 2} {c, b, p : 1}	c : 3, f : 2, a : 2, m : 2, b : 1	{c, p : 3}
{m}	{c, f, a, m : 2} {c, f, a, b, m : 1}	c : 3, f : 3, a : 3, b : 1	{c, m : 3}, {f, m : {a, m : 3}, {c, m {c, a, m : 3}, {a, f, {a, c, f, m : 3}
{b}	{c, f, a, b : 1} {f, b : 1} {c, b : 1}	c : 2, f : 2, a = 1,	—
{a}	{c, f, a : 3}	c : 3, f : 3, cf : 3 ~~a : 3~~	{c, a : 3}, {f, a : 3} {c, f, a : 3}
{f}	{c, f : 3} {f : 1}	c : 3	{f, c : 3}
{c}	{c : 4}	c : 4	

{m, c, a, f : 3}

m → {c, a, f}

m, c → {a, f}

ca → {m, f}

mf → {c, a}

$c \to \{m,a,f\}$, $a \to \{m,c,f\}$, $f \to \{m,c,a\}$

$cf \to \{m,a\}$, $af \to \{m,c\}$, ~~$ma \to \{c,f\}$~~

$ma \to \{c,f\}$, $m,c,a \to f$, $m,a,f \to c$,

$m,c,f \to a$

find Confidence value for $m \to c,a,f$

$$Conf(m \to c,a,f) = \frac{Suppcount(m \cup c,a,f)}{Supp(m)}$$

$$\frac{3}{3} = 1$$

ie 100%.

Unit 1 :- How to calculate Pearson Coefficient.

It is usually used in data integration phase to detect redundancy for numerical values.

$$r = \frac{cov(x,y)}{\sigma x \cdot \sigma y}$$

X	y	$x - \bar{x}$	$y - \bar{y}$
1	2	-2	-4
2	4	-1	-2
3	6	0	0
4	8	1	2
5	10	2	4
$\bar{x} = 3$	$\bar{y} = 6$		

$\frac{1+2+3+4+5}{5}$ $\left(\frac{2+4+6+8+10}{5}\right)$

$$Cov(x,y) = \frac{1}{n}\sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{5}\left((-2)*(-4) + (-1)*(-2) + (0) + (1*2) + (2*4)\right)$$

$$= \frac{20}{5} = \underline{\underline{4}}$$

$$\sigma_x = \sqrt{\frac{1}{n}(x-\bar{x})^2}$$

$$= \sqrt{\frac{1}{5}(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}$$

$$= \sqrt{\frac{10}{5}2} = \underline{\underline{1.41}}$$

$$\sigma_y = \sqrt{\frac{1}{n}(y-\bar{y})^2}$$

$$= \sqrt{\frac{1}{5}(-4)^2 + (-2)^2 + 0(2)^2 + (4)^2}$$

$$= \sqrt{\frac{40}{5}8} = \underline{\underline{2.82}}$$

Pearson coeff $r = \dfrac{4}{(1.41)(2.82)} = \underline{\underline{1.00}}$

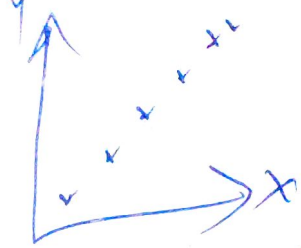
If $r = 1 \Rightarrow$ perfect +ve correlation between 2 variables.

As x increases, Y also increases.

→ The relationship is perfectly linear.

→ All data points lie exactly on straight line

with the slope.



So they are redundant so one can be removed during feature selection or data reduction.

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Unit 2 Contd. - -

Apriori Method :- Limitations :-

We have another method :- 1) Hash-based technique → reduces size of candidate k-itemsets, C_k for $k > 1$.

2) Partioning :- partitioning data to find candidate itemsets.
3) Sampling - mining on a subset of data.

min sup = 30%
each partition contains 3 transactions

TID	Items
T_1	
T_2	
T_3	
T_4	
T_5	
T_6	
T_7	
T_8	A, C, D
T_9	B, D

4) Dynamic itemset Counting.

Mining multilevel Association Rules.

" " - Dimensional Association

Multi-dimensional rule = ≥ 2 dimensions.

→ Categorical Attributes → finite no of values - Data cube approach

→ Quantitative Attributes - numeric → discretization clustering & gradient approach

Mining Quantitative Associations

→ static discretization

→ Dynamic "

→ Clustering

→ Deviation

Negative and Rare patterns.

↓ very low support but interesting

↓ infrequent patterns are more interesting than those that are frequent

Defining Negative Correlated Patterns

200 transactions. A has 100, B has 100.

Support of A : $S(A) = 100/200 = 0.5$

" " B : $S(B) = 100/200 = 0.5$

Joint support : $S(A \cup B) = 1/200 = 0.005$

If independent, expected support $= S(A) * S(B)$

$= 0.5 \times 0.5$

$= 0.25$

$S(A \cup B) < S(A) * S(B)$

This def^n for this :—

If x & y are frequent, but rarely occur together,
$sup(x \cup y) < sup(x) \cdot sup(y)$

Def^n 2 :— Constraint-based (Query-Directed) Mining

Constraints in Data Mining :—

→ Knowledge type
→ Data
→ Dimension / level
→ Rule (or pattern)
→ Interestingness

Meta-Rule Guided Mining :—

$P_1 \wedge P_2 \wedge \dots \wedge P_l = Q_1 \wedge Q_2 \wedge \dots \wedge Q_r$

where we search patterns using meta-rules (high level templates or constraints that describes the kind of rules a user is interested in finding

eg of meta rule :- Find associations where a customer buying a beverage is also likely to buy a snack

Now the mining system searches for rules matching template, such as :—

→ Customers who buy coffee often buy cookies.
→ Customers who buy soft drink often buy chips.

Rather than discovering every possible association, it focus on user-defined structure

Method to find meta-rules

1. Domain Expert knowledge.
2. Data Generalization using Concept Hierarchies
3. Metadata Analysis
4. User-defined templates.
5. Attribute-Oriented Induction (AI)
6. Ontology-based approaches

Steps in Meta-Rule Generation :-

1. Identify relevant attributes
2. Build Concept hierarchies or ontologies
3. Generalize data to high-level concepts
4. Define rule templates (meta rules)
5. Apply mining algorithms to discover specific rules matching the templates
6. Evaluate rules using support, confidence, lift etc.

Constraint-Based Freq Pattern Mining - Process of mining frequent itemsets, association rules or sequential patterns under a set of constraints specified by the user.

Constraints ↓

* reduces search space.
* Improves mining efficiency
* produces more meaningful patterns.
* Incorporates user interests & domain knowledge.

Type of Constraints :-

1. Knowledge-type Constraints
2. Data Constraints
3. Dimension/Level Constraints
4. Rule Constraints
5. Aggregate constraints

Constraint categories based on pruning/pattern-space pruning constraints:-

1. Anti-monotonic Constraint
2. Monotonic Constraint
3. Succinct Constraint
4. Convertible Constraint

Pattern ~~Sp~~ Naive Algorithm

→ Apriori + Constraint
→ Not practical / not feasible for very large (colossal) patterns.

Colossal patterns:-

eg:- $T_1 = \{1, 2, 3, 4, \ldots, 39, 40\}$
$T_2 = \{1, 2, 3, 4, \ldots, 39, 40\}$
$\vdots$
$T_{10} = \{1, 2, 3, 4, \ldots, 39, 40\}$

Colossal patterns are extremely large frequent itemsets, making traditional Apriori-based approaches inefficient.

Mining Compressed Patterns:- δ clustering

Pattern distance Measure:

$$D(P_1, P_2) = 1 - \left| \frac{T(P_1) \cap T(P_2)}{T(P_1) \cup T(P_2)} \right|$$

This is based on Jaccard similarity, where:

$$J(P_1, P_2) = \left| \frac{T(P_1) \cap T(P_2)}{T(P_1) \cup T(P_2)} \right|$$

and the distance is :-

$$D = 1 - J$$

Jaccard Similarity Threshold (δ)

δ value	Meaning
0.2	Loose clustering
0.5	Moderate "
0.8	Very strict "

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Unit-3 Classification.

Base Classifier

- Decision tree based methods
- Rule based
- Nearest-Neighbor
- Naive Bayes & Bayesian Belief N/W
- SVM
- Neural N/W, Deep Neural N/W

Ensemble classifier

- Boosting, bagging, Random forests

Classification techniques

(1) Decision tree method. ⇒ There could be more than 1 tree that fits the same data. (ie 2^n trees)

Hunt's Algorithm.

Method. Test Cond^n for Nominal Attributes :-

- Multi-way split
- Binary split

Test Cond^n for ordinal attributes

Test cond^n for Continuous attributes.

Splitting based on Continuous attributes :-

How to determine the Best Split :-

Find the impurity

⇓

Greedy approach

Low degree of impurity ⟸ is preffered.

Measures of Node Impurity :-

① **Gini Index :-** $\text{gini index} = 1 - \sum_{i=0}^{c-1} p_i(t)^2$

where $p_i(t)$ is the freq of class i at node t.

② **Entropy :-**

$$\text{Entropy} = -\sum_{i=0}^{c-1} p_i(t) \log_2 p_i(t)$$

③ **Misclassification error :-**

$$\text{classification error} = 1 - \max[p_i(t)]$$

① Compute impurity measure (P) — parent before splitting

② " " " (M) — after splitting

child node

$$\boxed{\text{Gain} = P - M.}$$

Gini index for node t :-

$$\text{Gini index} = 1 - \sum_{i=0}^{c-1} p_i(t)^2$$

Explain decision tree construction hunts algorithm
Find best nodes for split - using gini value

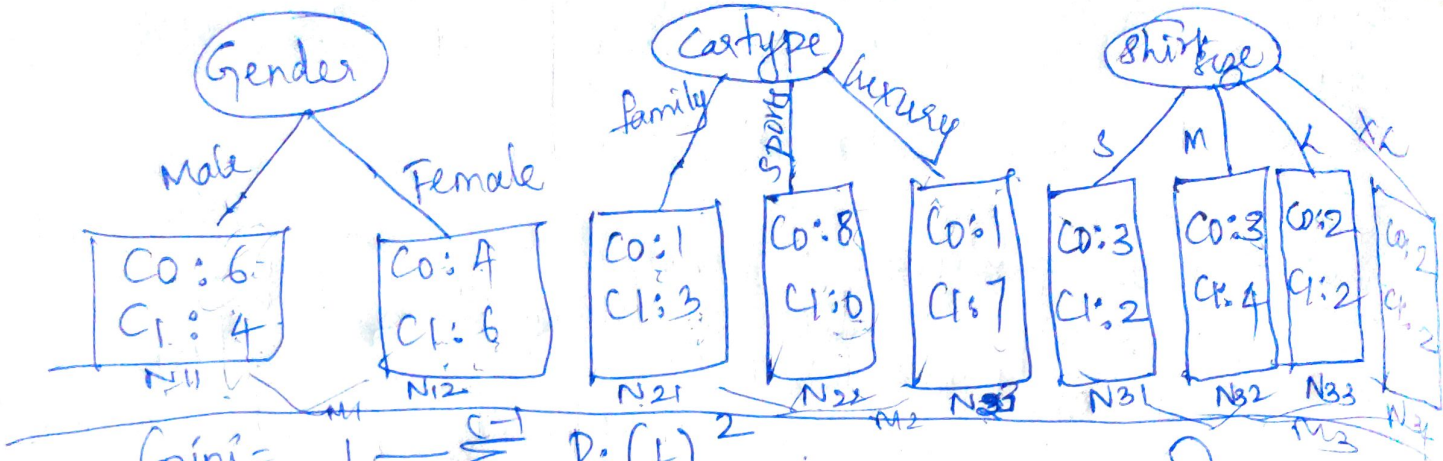
Decision tree based classification

Adv:- 1) Inexpensive to construct
2) Easy to interpret for small-sized trees
3) Robust to noise
4) extremely fast at classifying unknown reco
5) easily handle redundant attributes
6) " " irrelevant "

Disadv:- Each decision boundary involves singl attribute
- Greedy nature may cause favouritism.

Q1

Cust ID	Gender	Car Type	Shirt size	class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Ex Large	C0
6	M	Sports	Ex Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Ex Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Ex Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1



$$\text{Gini} = 1 - \sum_{i=0}^{C-1} P_i(t)^2$$

Gini value for P $= 1 - \left[\left(\frac{10}{20}\right)^2 + \left(\frac{10}{20}\right)^2\right]$

$= 1 - \left[(0.5)^2 + (0.5)^2\right]$

$= 1 - 0.5$

$= 0.5$

because C0: 10, C1: 10

Gini value for Gender

for N_{11} :- $1 - \left[\left(\frac{6}{10}\right)^2 + \left(\frac{4}{10}\right)^2\right] = 0.48$

$N_{12} = 1 - \left[\left(\frac{4}{10}\right)^2 + \left(\frac{6}{10}\right)^2\right] = 0.48$

Weighted Gini value of N_{11} and N_{12} = Gain of M_1 =

$$M_1 = \text{Gini}_{split} = \sum_{i=1}^{k} \frac{n_i}{n} \text{Gini}(i)$$

where, n_i – no of records in node

n – total no of records

where $i = 1 \& 2$ since 2 nodes

$= \frac{10}{20} * \text{Gain}(1) + \frac{10}{20} * \text{Gain}(2)$

$= \frac{10}{20} * 0.48 + \frac{10}{20} * 0.48.$

$M_1 = 0.24 + 0.24 = \boxed{0.48}$

Gini value for Cartype :-

$N_{21} :- 1 - \left[\left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}\right)^2\right] = 0.375$

$N_{22} = 1 - \left[\left(\frac{8}{8}\right)^2 + \left(\frac{0}{8}\right)^2\right] = 0$

$N_{23} = 1 - \left[\left(\frac{1}{8}\right)^2 + \left(\frac{7}{8}\right)^2\right] = 0.21875$

Weighted Gain of $N_{21}, N_{22}, N_{23} \Rightarrow M_2$

$M_2 = \frac{4}{20} * \text{Gain}(1) + \frac{8}{20} * \text{Gain}(2) + \frac{8}{20} * \text{Gain}(3)$

$= \frac{4}{20} * 0.375 + \frac{8}{20} * 0 + \frac{8}{20} * 0.21875$

$= 0.075 + 0 + 0.0875$

$\boxed{M_2 = 0.1625}$

Gini value for Shirt size :-

$N_{31} = 1 - \left[\left(\frac{3}{5}\right)^2 + \left(\frac{2}{5}\right)^2\right] = 0.48.$

$N_{32} = 1 - \left[\left(\frac{3}{7}\right)^2 + \left(\frac{4}{7}\right)^2\right] = 0.4897$

$N_{33} = 1 - \left[\left(\frac{2}{4}\right)^2 + \left(\frac{2}{4}\right)^2\right] = 0.5$

$$N_{34} = 1 - \left[\left(\frac{2}{4}\right)^2 + \left(\frac{2}{4}\right)^2\right] = 0.5$$

Weighted gain for $N_{31}, N_{321}, N_{331}, N_{34} = M_3$

$$M_3 = \frac{5}{20} * \text{Gain}(1) + \frac{7}{20} * \text{Gain}(2) + \frac{4}{20} * \text{Gain}(3) + \frac{4}{20} * \text{Gain}(4)$$

$$= \frac{5}{20} * (0.48) + \frac{7}{20} * (0.4897) + \frac{4}{20} * 0.5 + \frac{4}{20} * 0.5$$

$$= (0.25)(0.48) + (0.35) * (0.4897) + (0.2) * (0.5) + (0.2) * (0.5)$$

$$= 0.12 + 0.171395 + 0.1 + 0.1$$

$$\boxed{M_3 = 0.4913}$$

Gain $= P - M_1 \Rightarrow 0.5 - 0.48 = 0.02$

$= P - M_2 \Rightarrow 0.5 - 0.1625 = 0.3375$

$= P - M_3 \Rightarrow 0.5 - 0.4913 = 0.0087$

0.02 (M1) vs 0.3375 (M2) vs 0.0087 (M3) → Less gain

So M_3 is considered as root node.

Entropy: $-\sum_{i=0}^{c-1} P_i(t) \cdot \log_2 P_i(t)$

Entropy of $N_{11} = -\left[\frac{6}{10} * \log_2\left(\frac{6}{10}\right) + \frac{4}{10} * \log_2\left(\frac{4}{10}\right)\right]$

$= -[-0.2922] = +0.2922$

Entropy of $N_{12} = -\left[\frac{4}{10} * \log_2\left(\frac{4}{10}\right) + \frac{6}{10} * \log_2\left(\frac{6}{10}\right)\right]$

$= 0.2922$

weighted gain of M_1

$= \frac{10}{20} * (0.2922) + \frac{10}{20} * (0.2922)$

$M_1 = 0.2922$

Entropy of $N_{21} = -\left[\left(\frac{1}{4}\right) \log_2\left(\frac{1}{4}\right) + \frac{3}{4} * \log_2\left(\frac{3}{4}\right)\right.$

$= 0.244$

$N_{22} = -\left[\left(\frac{8}{8}\right) \log_2\left(\frac{8}{8}\right) + \left(\frac{0}{8}\right) * \log_2\left(\frac{0}{8}\right)\right]$

$= 0$

$N_{23} = -\left[\left(\frac{1}{8}\right) \log_2\left(\frac{1}{8}\right) + \left(\frac{7}{8}\right) * \log_2\left(\frac{7}{8}\right)\right]$

$= 0.1636.$

weighted gain of $M_2 = \frac{4}{20} * 0.24 + \frac{8}{20} * 0 + \frac{8}{20} * 0.1636$

$M_2 = 0.11424$

Entropy of $N_{31} = -\left[\left(\frac{3}{5}\right)\cdot\log_2\left(\frac{3}{5}\right) + \left(\frac{2}{5}\right)\log_2\left(\frac{2}{5}\right)\right]$

$= +0.2922$

Entropy of $N_{32} = -\left[\frac{3}{7}\cdot\log_2\left(\frac{3}{7}\right) + \left(\frac{4}{7}\right)\log_2\left(\frac{4}{7}\right)\right]$

$= 0.2965$

Entropy of $N_{33} = -\left[\frac{2}{4}\cdot\log_2\left(\frac{2}{4}\right) + \left(\frac{2}{4}\right)\log_2\left(\frac{2}{4}\right)\right]$

$= 0.3010$

Entropy of $N_{34} = -\left(\frac{2}{4}\cdot\log_2\left(\frac{2}{4}\right) + \left(\frac{2}{4}\right)\log_2\left(\frac{2}{4}\right)\right)$

$= 0.3010$

Weighted gain of $M_3 = \left(\frac{5}{20}\right) * 0.2922 + \left(\frac{7}{20}\right) * 0.2965$

$+ \left(\frac{4}{20}\right) * (0.3010) + \left(\frac{4}{20}\right) * 0.3010.$

$= 0.07305 + 0.103775 + 0.0602$

$+ 0.0602$

$M_3 = 0.297225$

Gain $= P - M$

Entropy at P ⟨ $C_0 : 10$, $C_1 : 10$

Entropy for P = $-\left[\left(\frac{10}{20}\right)\log_2\left(\frac{10}{20}\right) + \left(\frac{10}{20}\right)\log_2\left(\frac{10}{20}\right)\right]$

$= -\left[(0.5)\log_2(0.5) + (0.5)\log_2(0.5)\right]$

$= \underline{0.3010}$

Gain:- $P - M_1 \Rightarrow 0.3010 - 0.2922 = 0.0088$

$P - M_2 \Rightarrow 0.3010 - 0.11424 = 0.1867$

$P - M_3 \Rightarrow 0.3010 - 0.297225 = 0.003775$

$= 0.003775$

from this M_1 Vs M_2 Vs M_3

M_3 is smallest ⇒ Consider this as root node

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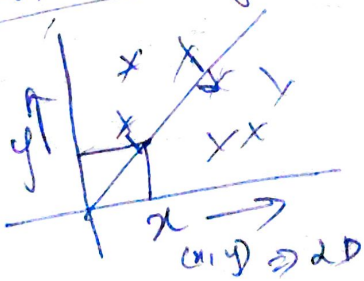
Linear Regression - Supervised → x & y.

future prediction purpose



$y = w_0 + w_1 x$ → slope.

↓ intercept

$y(x, w) = w_0 + w_1 x_2 + w_2 x_2 + w_3 x_3 + \ldots + w_n x_n$

or

$y(x, w) = w_0 + \sum_{i=1}^{m} w_i \theta_i(x)$

$\theta_i(x)$ → learning funct^n.

Basis funct^n - $\theta_j(x) = e^{\left\{\frac{(x-M)^2}{2\sigma^2}\right\}}$ - Gaussian distribution

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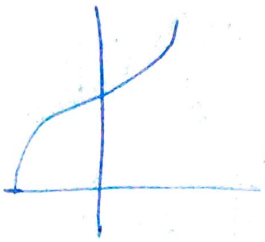
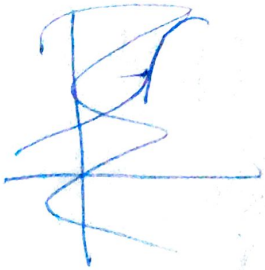
Sigmoid funct^n :-





Sigmoid funct^n $= \frac{1}{1+e^{-x}}$

These funct^n are used to calc w values i.e w_0, w_1

$$\theta_j(x) = \sigma\left(\frac{x-\mu_j}{\sigma}\right)$$

$$\tanh(x) = 2\sigma(x) - 1$$

Least Square Regression:-

$$\text{Intercept}(w_0) = \frac{(\Sigma y - w_1(\Sigma x))}{N}$$

N - No of datapts.
w_0 - Intercept
w_1 - slope.

$$\text{slope}(w_1) = \frac{(N\Sigma xy - (\Sigma x)(\Sigma y))}{(N\Sigma x^2 - (\Sigma x)^2)}$$

eg: $x = 60, 61, 62, 63, 65.$

$y = 3.1, 3.6, 3.8, 4.0, 4.1$

x	y
60	3.1
61	3.6
62	3.8
63	4.0
65	4.1

$N = 5$

$\Sigma x^2 = 60^2 + 61^2 + 62^2 + 63^2 + 65^2 = 19359$

$\Sigma y^2 = (3.1)^2 + (3.6)^2 + (3.8)^2 + 4^2 + (4.1)^2 = 69.82$

Σxy ~~w_1~~ = ~~5.~~ $[(60*3.1)+(61*3.6)+(62*3.8)+(63*4)+(65*4.1)]$ ~~cm~~

$= 1159.7$

$\Sigma x = 60+61+62+63+65 = 311$

$\Sigma y = 3.1+3.6+3.8+4+4.1 = 18.6$

$\therefore w_1 = \dfrac{5(1159.7) - (311)(18.6)}{5.(19359) - (311)^2}$

$w_1 = \dfrac{13.9}{74} = 0.187$ ← slope

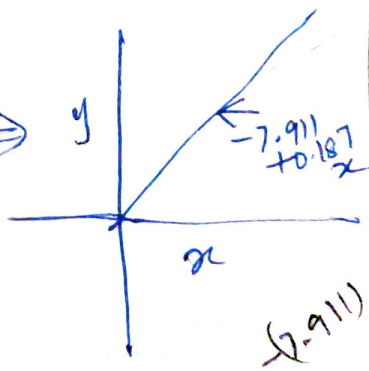
$w_0 = \dfrac{(\Sigma y - w_1(\Sigma x))}{N}$

$= \dfrac{18.6 - (0.187)(311)}{5}$

$= -7.911$ or ≈ -8.098

↑ intercept

when so $y = -7.911 + 0.187x$ ⇒



when $x = 67$, $y = ?$

$= -7.911 + 0.187(67)$

$= 4.618$

$y = \underbrace{0.187x}_{mx} + \underbrace{(-7.911)}_{c}$

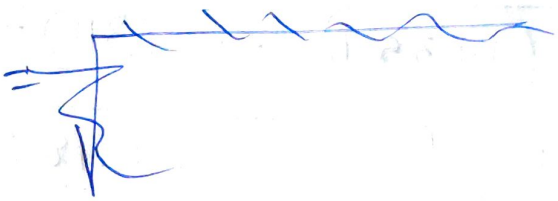
When $x = 60$, $y = 3.3$

But in 8, $y = 3.1$ for $x = 60$.

There is 0.2 difference.

Residual Error $\epsilon = \sqrt{\frac{\Sigma (y - y_{est})^2}{n-2}}$

$y_{est} \Rightarrow$ y estimation



Accurate value:-

$y = w_0 + w_1 x + \dots w_n x - \epsilon$

When $x = 60$, $y_{est}(60) = -7.911 + 0.187 * (60)$

$= 3.3$

$y_{est}(61) = -7.911 + 0.187(61)$

$= 3.4896$

$y(62) = -7.91 + 0.187(62)$

$= 3.684$

$y(63) = -7.91 + 0.187(63)$

$= 3.871$

$y(65) = -7.91 + 0.187(65)$

$= 4.245$

$(y - y_{est})^2$

$= (3.1 - 3.3)^2 + (3.6 - 3.49)^2 + (3.8 - 3.68)^2$
$+ (4 - 3.87)^2 + (4.1 - 4.2)^2$

$= 0.04 + 0.0121 + 0.0144 +$
$0.0169 + 0.0196.$

$= \underline{0.103}$

$\sqrt{\frac{\Sigma (y - y_{est})^2}{n-2}} = \sqrt{\frac{0.103}{3}} = \sqrt{0.03433} = \underline{0.185}$

$\varepsilon = \underline{0.185}$

$y = w_0 + w_1 x + \cdots w_n x - \varepsilon$

eg:- for $y_1 =$ 3.3 (y_{1est}) $-$ 0.185 (ε) $=$ 3.11 $\Rightarrow$ we got value in the table.

Logistic Regression :- — used for classification
↓ value b/w (0 & 1)
uses sigmoid funct^n.

$h_\theta(x) = \theta(x)$ → model parameter or slope function.

↓ hypothesis function.

eg for cancer &

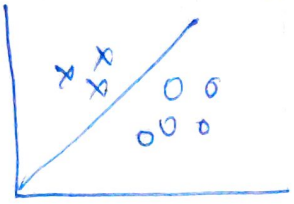
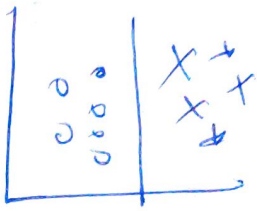
If threshold = 0.5.

$h_\theta(x) \geq 0.5$, $y = 1$ — malign (cancer cell)

$h_\theta(x) < 0.5$, $y = 0$ — Non malignant (not cancer cell)

Tasks to do :-
→ Logistic funct^n
→ Learn coefficients
→ Predict class labels

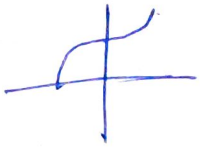
If it is possible to linearly seperable data only then logistic regression is used.


or

or

Logistic funct^n used :- Sigmoid funct^n :-

$$h_\theta(x) = \frac{1}{1+e^{-x}}$$, value lies b/w (0, 1)

When we calculate x & y values using sigmoid funct^n, we get sigmoid curve



eg:-

tumor size (x_1)	tumor volume (x_2)	y
2.7	2.5	0
1.4	2.3	0
3.3	4.4	0
3.06	3.05	0
5.3	2.75	1

Here $z = b_0 + b_1 x_1 + b_2 x_2$

So formula is :- $\frac{1}{1+e^{-(b_0+b_1x+b_2x_2)}}$

$$b = b + \alpha * (y - y_{pred}) * y_{pred} * (1 - y_{pred}) * x$$

when b_0, $x = 1$

$b_1 \Rightarrow x_2 = x_1$

$b_2 \Rightarrow x_2 = x_2$

We use stochastic gradient descent method

$x_1 = 2.7$, $x_2 = 2.5$

Initially $b_1, b_2, b_0 = 0$ in gradient descent

We will find proper b_0, b_1, b_2

$P(y)$

$$= \frac{1}{1+e^{-(0+0*x_1+0*x_2)}}$$

$$= \frac{1}{1+e^{-(0+0+0)}} = \frac{1}{1+e^{-0}} = \frac{1}{2} = 0.5$$

$P(0/1)$ ⇒ predicting whether 0 or 1

$y_{pred} = 0.5$, but $y = 0$.

Actual value is less than 0.5

if $h_\theta(x) \geq 0.5$, $y = 1$

Learning rate :- $\alpha = 0.3$

$$b_0 = 0 + (0.3).(0 - 0.5) * (0.5) * (1 - 0.5) * 1$$

$$= -0.0375$$

$$b_1 = 0 + (0.3)(0 - 0.5) * (0.5) * (1 - 0.5) * 2.7$$

$$= -0.1012$$

$$b_2 = 0 + (0.3)(0-0.5) * 0.5(1-0.5) * 2.5$$
$$= -0.09375$$

New $b_0 = -0.0375$
" $b_1 = -0.101$
" $b_2 = -0.09375$

} — Epoch - 1.

2nd record :- $x_1 = 1.4$, $x_2 = 2.3$

$$P(0/1) = \frac{1}{1+e^{-(-0.0375 + (-0.101 * 1.4) + (-0.09375 * 2.3))}}$$

$$= \frac{1}{1+e^{0.3945}} = \frac{1}{1+1.48} = \frac{1}{2.48}$$
$$= 0.403$$

$y_{pred} = 0.403 < 0.5$
$\therefore y = 0$
even in the table, $y = 0$.

22/6/26.

Bayes classifier

Conditional probability :—

$$P(Y/X) = \frac{P(X,Y)}{P(X)}$$

$$P(X/Y) = \frac{P(X,Y)}{P(Y)}$$

Bayes theorem :— $$P(Y/X) = \frac{P(X/Y) \cdot P(Y)}{P(X)}$$

evidence

For continuous value,

Normal distribution :-

$$P(X_i \mid Y_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} \cdot e^{-\frac{(X_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

- one for each (X_i, Y_i) pair.

Direct probability formula can't be used for continuous value (numerical value).

For nominal / string data, we can use probability formula.

Q.1 :-

ID	Refund	Marital Status	Taxable Income	Evade
1.	Yes	Single	125k	No
2.	No	Married	100k	No
3.	No	Single	70k	No
4.	Yes	Married	120k	No
5.	No	Divorced	95k	Yes
6.	No	Married	60k	No
7.	Yes	Divorced	220k	No
8.	No	Single	85k	Yes
9.	No	Married	75k	No
10.	No	Single	90k	Yes

$X = (\text{Refund} = \text{No}, \text{Marital Status} = \text{Divorced}, \text{Tax-income} = 120k)$

$P(Y \mid X) = ?$

$P(Y \mid X) = P(\text{Refund} = \text{No}, \text{marital Status} = \text{Divorced}, \text{Tax income} = 120 \mid Y) \cdot P(Y)$

$$= \left[P\left(\text{refund} = \text{no} \mid Y\right) \cdot P\left(\text{marital status} = \text{Divorced} \mid Y\right) \cdot P\left(\text{Taxable income} = 120k \mid Y\right) \right] \cdot P(Y)$$

Prior Probabilities: — for Refund = NO, Yes

$P(NO) = 7/10$

$P(Yes) = 3/10$

$P(\text{refund} = NO / Yes) = 3/3 = 1$

$P(\text{refund} = NO / NO) = 4/7$

$P(\text{refund} = Yes / Yes) = 0/3 = 0$

$P(\text{refund} = Yes / NO) = 3/7$

for marital status = Divorced, Single, Married

$P(\text{marital status} = \text{Divorced} / Yes) = 1/3$

$P(\text{marital status} = \text{Married} / Yes) = 0/3 = 0$

$P(\text{marital status} = \text{Single} / Yes) = 2/3$

$P(MS = \text{Divorced} / NO) = 1/7$

$P(MS = \text{Married} / NO) = 4/7$

$P(MS = \text{Single} / NO) = 2/7$

for ... (... numerical data)

$P(\text{Tax Inc}/\text{No}) = \frac{2}{125}$

100
70
120
60
220
75

7 record.

Total.

770 ~~5~~

$$\boxed{\mu_{no} = \frac{770}{7} = 110}$$ Since for No, n = 7

$$\sigma^2_{no} = \frac{1}{n} \sum (x_i - \mu_i)$$

$$= \frac{1}{n-1} \sum (x_i - \mu_i)^2$$

$$= \frac{1}{(7-1)} \left((125-110)^2 + (100-110)^2 + (70-110)^2 + (120-110)^2 + (60-110)^2 + (220-110)^2 + (75-110)^2 \right)$$

$$= \frac{225 + 100 + 1600 + 100 + 2500 + 12100 + 1225}{7}$$

$$\sigma^2 = 2975 \cdot \frac{17850}{(7-1)} = \frac{2975}{2550}$$

$$\sigma = \sqrt{2975} = 54.54$$

$$P(x_i/y) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{\frac{-(x_i-\mu)}{2\sigma^2}}$$

$$P\left(\frac{\text{tax}}{\text{income}}/\text{NO}\right) = \frac{1}{\sqrt{2.\pi.(2975)}} \cdot e^{\frac{-(120-110)^2}{2.(2975)}}$$

$$= \frac{1}{\sqrt{18692.47}} \cdot e^{\frac{-(100)^{50}}{2.2975}}$$

$$\frac{1}{136.72} \cdot (2.17)^{\frac{-50}{2975}}$$

$$= 0.0072$$

P (Tax-Income/Yes) = 95, 85, 90

$$270 \qquad \mu = \text{or } \bar{x} = \frac{270}{3} = 90$$

$$\sigma^2 = \frac{1}{n-1}\left[(95-90)^2 + (85-90)^2 + (90-90)^2\right]$$

$$= \frac{1}{2}\left[5^2 + 5^2 + 0\right]$$

$$\frac{25+25}{2} = 25$$

$$\sigma = \sqrt{25} = 5$$

$$P(\text{tax-income}/\text{yes}) = \frac{1}{\sqrt{2\pi.(25)}} \cdot e^{\frac{-(120-90)^2}{2(25)}}$$

$$= \frac{1}{\sqrt{157.07}} \cdot e^{-\frac{30^2}{2*25}} \qquad = \frac{1}{} \cdot 0.606 =$$

$X = (\text{Refund} = \text{no}, \text{MS} = \text{Divorced}, \text{Tax, Income} = 120)$

$P(\text{ref} = \text{no}/\text{No}) * P(\text{MS} = \text{div}/\text{No}) * P(\text{Tax Inco} = 120/\text{No})$

$= \left(\frac{4}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot 0.0072 = 0.000587$

$P(X|\text{yes}) = 1 * \frac{1}{3} * 10^{-9} = 0.33 \times 10^{-9}$

$P(y/x) = \frac{P(x/y) * P(y)}{P(x)}$

$P(\text{No}/x) = \frac{P(x/\text{No}) * P(\text{No})}{P(x)}$

$= 0.000587 * 7/10 = 0.00041$

$P(\text{Yes}/x) = P(x/\text{Yes}) * P(\text{Yes})$

$= 0.33 * 10^{-9} * 3/10.$

$= 9.9 * 10^{-11}$

If $P(\text{No}/x) > P(\text{Yes}/x)$

$0.00041 > 9.9 \times 10^{-11}$ ✓ True.

Class label assigned to $X = \text{No}$.

ie Refund = No, Marital Status = Divorced, Tax-income = No.

for $X = (\text{refund} = \text{yes}, \text{Mar} = \text{married}, \text{Tax-income} = 95)$

~~(refund = yes/No) = 3/7, refund = (~~

for tax-income : 95 :-

$P(\text{tax-income}/\text{no}) =$

125
100
70
120
60
220
75

770.

$\mu = \frac{770}{7} = 110$

$\sigma^2 = 2975, \quad \sigma = 54.54$

$$P(\text{tax}/\text{No}) = \frac{1}{\sqrt{2.\pi.(2975)}} \cdot e^{-\frac{(95-110)^2}{2*2975}}$$

$$= \frac{1}{136.72} \cdot e^{-\frac{225}{2*2975}}$$

$$= 0.00731 * e^{-\frac{225}{5950}}$$

$$= 0.00731 * .e^{-0.0378}$$

$$= 0.00731 * 0.9629$$

$$= 0.00703$$

$P(\text{tax-income/yes}) = \begin{matrix} 95 \\ 85 \\ 90 \\ \hline 270 \end{matrix}$ $\quad \mu = 270/3 = 90.$

$\sigma^2 = 25, \quad \sigma = 5.$

$$P(\text{tax/yes}) = \frac{1}{\sqrt{2 \cdot \pi \cdot (25)}} \cdot e^{-\frac{(95-110)^2}{2 \cdot 25}}$$

$$\frac{1}{\sqrt{157.07}} \cdot e^{-4.5}$$

$$\frac{1}{12.53} * 0.011108$$

$$= 0.000886$$

$$P(X/\text{yes}) = P(\text{ref}=Y/\text{yes}) \cdot P(MS=\text{married}/\text{yes}) \cdot P(T.\text{Inco}=95/\text{yes})$$

$$= 0 \cdot 0 \cdot 0.000886 = 0$$

$$P(\text{yes}/X) = P(X/\text{yes}) \cdot P(\text{yes})$$

$$= 0 \cdot 0.3 = 0$$

$$P(X/NO) = P(\text{ref}=Y/NO) \cdot P(MS=\text{married}/NO) \cdot P(T\text{-Income}=95/NO)$$

$$= 3/7 \cdot 4/7 \cdot 0.00703$$

$$= 0.00401$$

$$P(NO/X) = P(X/NO) \cdot P(NO) = 0.00401 * 0.7$$

$$= 0.0028$$

Since $P(No/x) > P(Yes/x)$

class label = NO

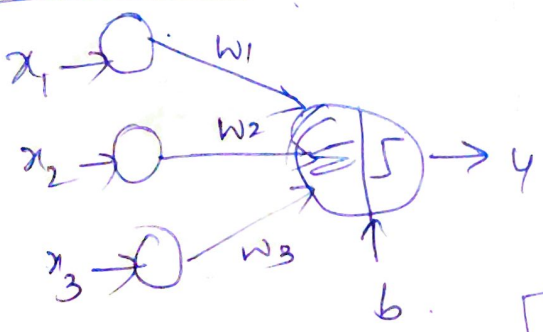
23/6/26

Artificial Neural Networks (ANN)

↓

Collection of simple processing units (nodes) that are connected by directed links (edges).

Perceptron architecture (single neuron)



$$y = \begin{cases} 1, & \text{if } w^T x + b = 0 \\ -1, & \text{otherwise} \end{cases}$$

$$\hat{y} = \text{sign}(\tilde{w}^T \tilde{x})$$

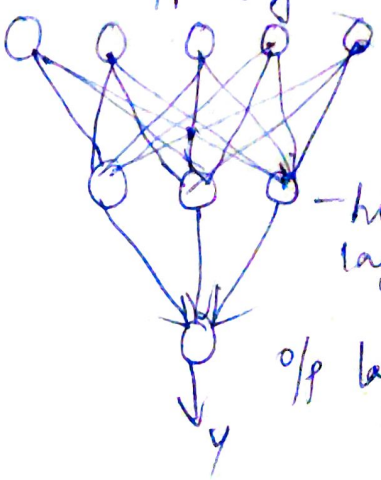
where $\tilde{w} = (w^T b)^T$

$\tilde{x} = (x^T 1)^T$

It is related to logistic regression. But the activation function is sign instead of sigmoid.

Multi-layer neural N/W

– More than one hidden layer of computing nodes



→ Every node in this layer transmits activations forward to nodes of next layer.

→ Its also called "feed-forward neural networks"

network output :- $x_1w_1 + x_2 + x_3w_3 + \dots - x_nw_n + b$.

outn_1 → sigmoid :- $\frac{1}{1+e^{-x}}$

y = network o/p ie $(x_1w_1 + x_2w_2 + \dots + b)$

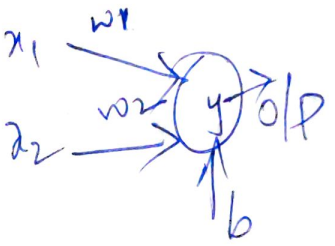
So outn_1 = $\frac{1}{1+e^{-(x_1w_1 + x_2w_2 + b)}}$

$y = \begin{cases} 1 & \geq 0 \\ 0 & < 0 \end{cases}$

AND gate funct^n

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

⇒



If $w_1 = 1$, $w_2 = 1$, $b = -1.5$

$x_1w_1 + x_2w_2 + b$.

Then $y = (0)(1) + (0)(1) + (-1.5)$

$= -1.5 \approx 0$

This is how weights are adjusted.

$= (0)(1) + (1)(1) + (-1.5)$

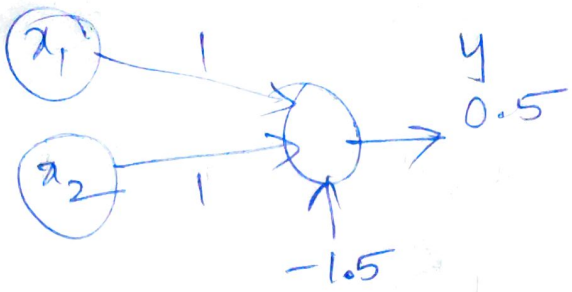
$1 - 1.5 = -0.5 \approx 0$

$= (1)(0) + (0)(1) + (-1.5)$

$1 - 1.5 = -0.5 \approx 0$

$= (1)(1) + (1)(1) - 1.5$

$2 - 1.5 = 0.5 \approx 1$



for OR Gate

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

ex:-

$w_1 = 1,\ w_2 = 1,\ b = -0.5$

$y = w_1 x_1 + w_2 x_2 + b$

$= (1)(0) + (1)(0) - 0.5$

$= -0.5 < 0 = 0$

$= (1)(0) + (1)(1) - 0.5$

$1 - 0.5 = 0.5 \approx 1$

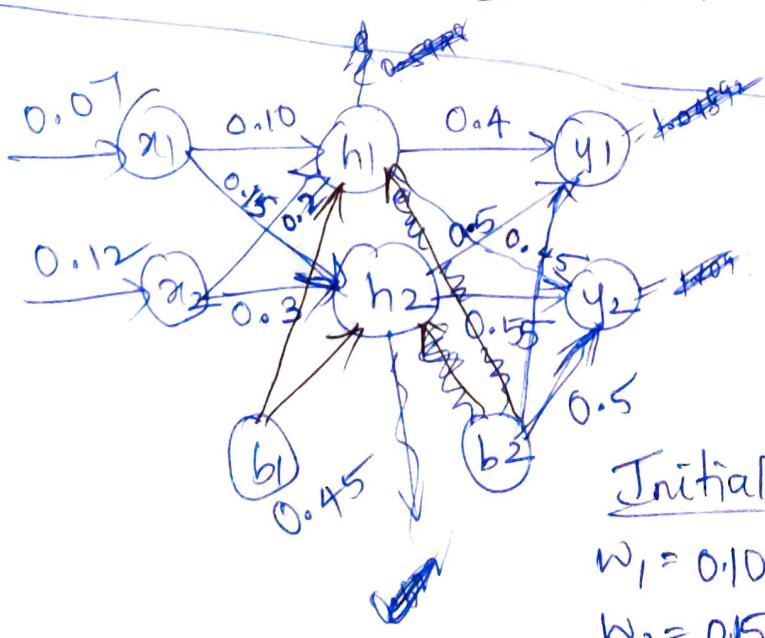
$= (1)(1) + (1)(0) - 0.5$

$1 - 0.5 = 0.5 \approx 1$

$= (1)(1) + (1)(1) - 0.5$

$2 - 0.5 \approx 1.5$

Q.



$x_1 = 0.07$

$x_2 = 0.12$

$b_1 = 0.45$

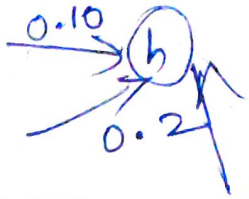
$b_2 = 0.5$

Initial weights

$w_1 = 0.10$	$w_5 = 0.40$
$w_2 = 0.15$	$w_6 = 0.45$
$w_3 = 0.20$	$w_7 = 0.5$
$w_4 = 0.30$	$w_8 = 0.55$

o/p of H_1 = ?

o/p of H_2 = ?



$$\text{net}_{H_1} = x_1 w_1 + x_2 w_2 + b$$

$$= 0.10\,(0.10) + (0.12)(0.2) + 0.45$$

$$= \underline{0.481}$$

$$\text{out}_{h_1} = \frac{1}{1+e^{\text{net}_{H_1}}} = \frac{1}{1+e^{-0.481}} = \underline{0.5948}$$

$$\text{net}_{H_2} = x_1 w_2 + x_2 w_4 + b$$

$$= (0.07)(0.15) + (0.12)(0.3) + 0.45$$

$$= \underline{0.4965}$$

$$\text{out}_{h_2} = \frac{1}{1+e^{\text{net}_{H_2}}} = \frac{1}{1+e^{-0.4965}} = \underline{0.622}$$

---

o/p of y_1 = ? , o/p of y_2 = ?

$$\text{net}_{y_1} = h_1 \text{o/p} \cdot (0.4) + h_2 \text{o/p} (0.5) + 0.5$$

$$= (0.5948)(0.4) + (0.622)(0.5) + 0.5$$

$$= \underline{1.04892}$$

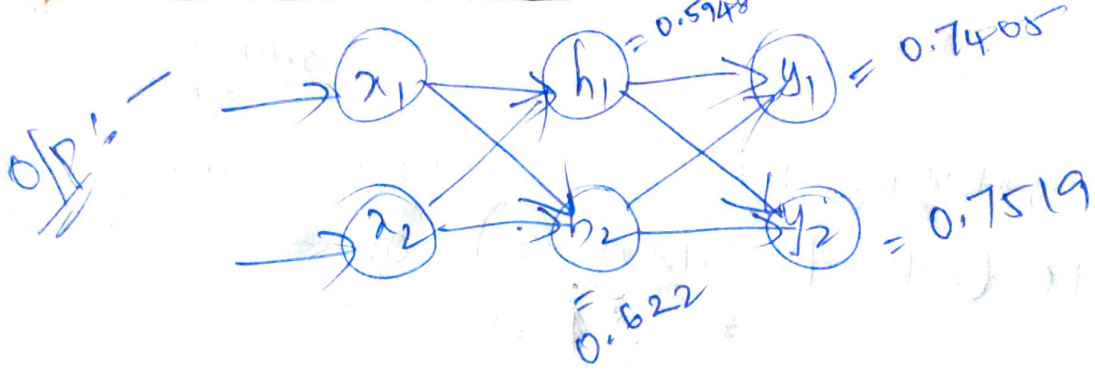
$$\text{net}_{y_2} = h_1 \text{o/p} \cdot (0.45) + h_2 \text{o/p} (0.55) + 0.5$$

$$= (0.5948)(0.45) + (0.622)(0.55) + 0.5$$

$$= \underline{1.109}$$

$$\text{out}_{y_1} = \frac{1}{1+e^{-1.04892}} = \underline{0.7405}$$

$$\text{out}_{y_2} = \frac{1}{1+e^{-1.109}} = \underline{0.7519}$$



Given : $T_1 = 0.20$, $T_2 = 0.95$

But $y_1 = 0.7405$, $y_2 = 0.7519$.

We shld calculate error b/w them

Total error = $\sum \frac{1}{2}(\text{target} - \text{output})^2$

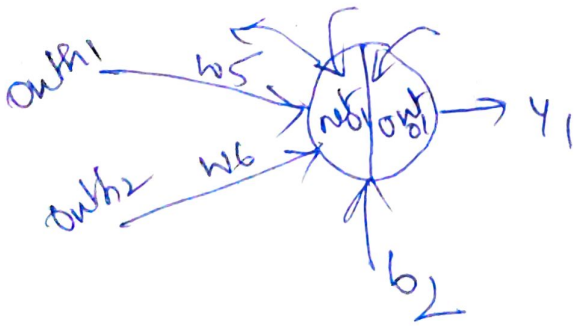
target = T_1 & T_2

o/p = out_{y_1}, out_{y_2}

Total error = $\frac{1}{2}\left[(0.2 - 0.7405)^2 + (0.95 - 0.7519)^2\right]$

$= \frac{1}{2}\left[0.2916 + 0.03924\right] = 0.1654$

Back propogation in ANN



$$\frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial w_5}$$

$E_{Total} = \frac{1}{2}\sum(target - out)^2$

Since there is only y_1,

we only take:- $\frac{1}{2}(target - out_1)^2$

$(target - out_2)^2$ rejected as we have only y_1

$2x = n \cdot x^{n-1}$

$2x$

$\frac{\partial E_{Total}}{\partial out_{o1}} = \frac{1}{2}(target - out_{o1})^2$ → after differentiation

$= 2(target - out_{o1})$

~~$\frac{\partial E_{Total}}{\partial out_{o1}} = target - out_{o1}$~~

After differentiation of $(target - out)^2$

$= 2.(target - out_{o1})$

Substitute in eqn:-

$\frac{\partial E_{Total}}{\partial out_{o1}} = \frac{1}{2} \cdot 2(target - out_{o1})$

$\therefore \boxed{\frac{\partial E_{Total}}{\partial out_{o1}} = target - out_{o1}}$

$\frac{\partial out_{o1}}{\partial net_{o1}} = \frac{1}{1+e^{-x}} \approx \frac{1}{1+e^{-net_{out}}}$

Partial differentiation of logistic function =

$= o/p * (1 - output)$.

$\frac{\partial out_{o1}}{\partial net_{o1}} = out_{o1} * (out_{o1} - 1)$

$\frac{\partial net_{o1}}{\partial w_5} = w_5 * out_{h1} + w_6 * out_{h2} + b$

this term doesn't consist for net_{o1} (network of 1)

$= w_5 * out_{h1}$

Consider w_5^5 in the form of x^0.

$$\boxed{w_5' = 1 . w_5^0}$$

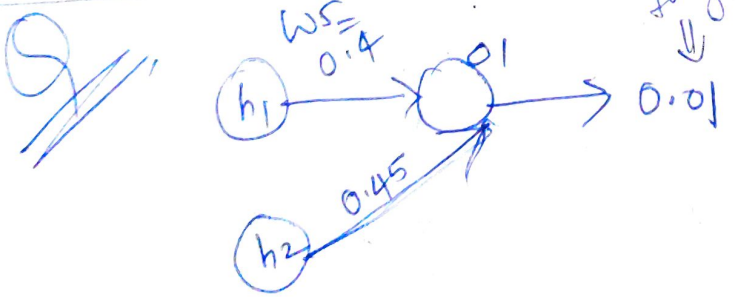
$= w_5^0 * out_{oh1} = out_{oh1}$

$\frac{\partial E_{total}}{\partial w_5} = (out_{o1} - target) * (out_{o1}) * (out_{o1} - 1) * out_{oh1}$

Updated weight calculation:—

$w_{5new} = w_5 - \eta * \frac{\partial E_{total}}{\partial w_5}$

η — learning rate = 0.5



$out_{h1} = 0.5932$

$out_{h2} = 0.5968$

$out_{o1} = 0.7513$

$out_{o2} = 0.7729$

$$E_{o1} = \frac{1}{2}(\text{target} - \text{out}_{o1})^2$$

$$= \frac{1}{2}(0.01 - 0.7513)^2$$

$$= 0.2748$$

$$\frac{\partial E_{Total}}{\partial w_5} = (\text{target} - \text{out}_{o1}) * \text{out}_{o1}(1 - \text{out}_{o1}) * \text{out}_{h_1}$$

$$= (0.01 - 0.7513) * (0.7513) * (1 - 0.7513) * 0.593$$

$$= 0.08216$$

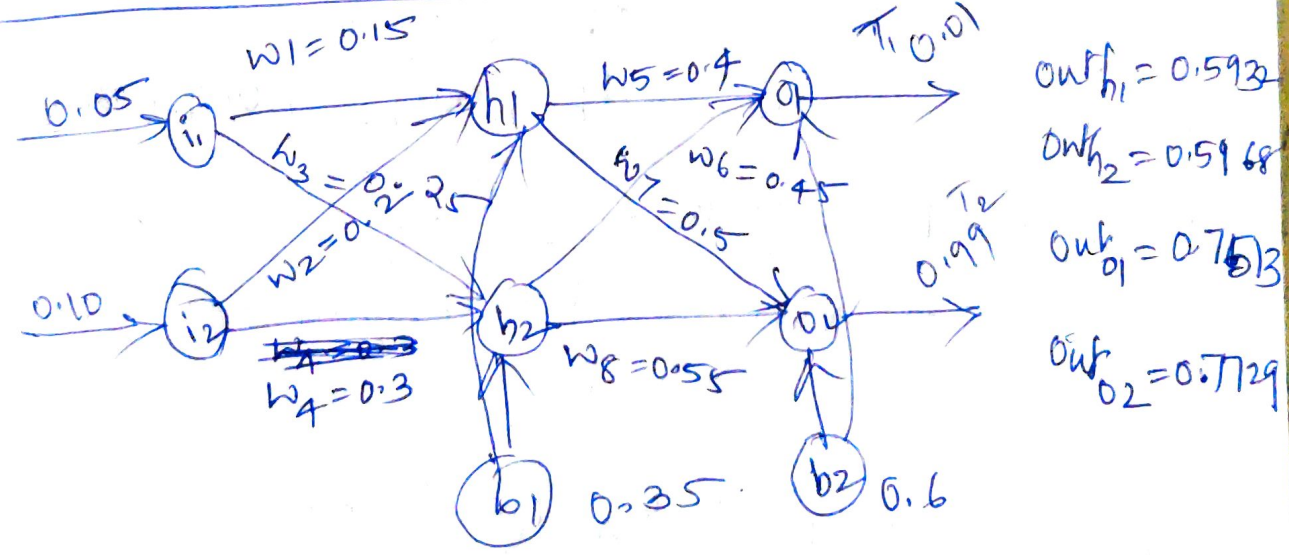
$$\text{Updated } w_5 \text{ value} = w_5 - \eta * \frac{\partial E_{Total}}{\partial w_5}$$

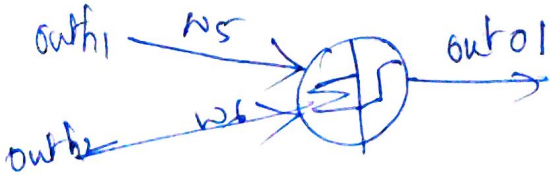
$$= 0.4 - (0.5 * 0.8216)$$

$$= 0.3589$$

Q, $\Rightarrow w_5 = 0.4$, but calculated, $w_5 = 0.3589$

E_{Total} should be approximately 0





Chuk of back propogation is required. & update weight

$E_{Total} = \frac{1}{2}\Sigma(target - out)^2$

Total error $= \frac{1}{2}\left[(0.01 - 0.7513)^2 + (0.99 - 0.7729)^2\right]$

$= \underline{0.2983} \neq 0$

∴ we should adjust weight.

$\frac{\partial E_{Total}}{\partial w_5} = (target - out_{o1})(out_{o1})(1 - out_{o1}) * out_{h1}$

$= (0.01 - 0.7513)(0.7513)(1 - 0.7513) * 0.5932$

$= \underline{0.082167}$

If out_{h1} is given, we can calculate this

But now, we should calculate output with only weights and T value.

we should calculate out_{h1}, out_{h2}, out_1, out_2.

$net_{h1} = w_1 \cdot i_1 + w_2 * i_2 + b_1$

$0.15 * (0.05) + (0.2)(0.10) + 0.35$

$= \underline{0.3775}$

$net_{h2} = w_3 \cdot i_1 + w_4 \cdot i_2 + b_1$

$= (0.25)(0.05) + (0.3)(0.10) + 0.35$

$= \underline{0.3925}$

$$out_{h_1} = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(0.3775)}} = \frac{1}{1+0.6855}$$

$$= \frac{1}{1.6855}$$

$$= 0.5932$$

$$out_{h_2} = \frac{1}{1+e^{-x}} = \frac{1}{1+e^{-(0.3925)}} = \frac{1}{1+0.6753}$$

$$= \frac{1}{1.6753}$$

$$= 0.5969$$

---

out_{o_1} :-

$$net_{o_1} = w_5 \cdot out_{h_1} + w_6 \cdot out_{h_2} + b_2$$

$$= (0.4)(0.5932) + (0.45)(0.569) + 0.6$$

$$= 1.09333$$

$$out_{o_1} = \frac{1}{1+e^{-(1.093)}} = \frac{1}{1+0.335}$$

$$= \frac{1}{1.335} = 0.749$$

out_{o_2} :- ?

$$net_{o_2} = w_7 \times out_{h_1} + w_8 \times out_{h_2} + b_2$$

$$= (0.5)(0.5932) + (0.55)(0.5969) + 0.6$$

$$= 1.2248$$

$$out_{o_2} = \frac{1}{1+e^{-(1.2248)}} = \frac{1}{1+0.2938}$$

$$= 0.7729$$

Total error :-

$$= \frac{1}{2}\left[\sum (\text{target} - \text{out})^2\right]$$

$$= \frac{1}{2}\left[(\text{target}_1 - \text{out}_{o1})^2 + (\text{target}_2 - \text{out}_{o2})^2\right]$$

$$= \frac{1}{2}\left[(0.01 - 0.7459)^2 + (0.99 - 0.7729)^2\right]$$

$$= \frac{[0.5415 + 0.04713]}{2}$$

$$= 0.2943 \neq 0$$

∴ weights shld be adjusted

$$\frac{\partial E_{Total}}{\partial w_5} = \frac{\partial E_{Total}}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial w_5}$$ → delta rule or chain rule

$$w_{new} = w - \eta \cdot \frac{\partial E_{Total}}{\partial w_5}$$

$$\frac{\partial E_{Total}}{\partial w_5} = (\text{target} - out_{o1}) * out_{o1} * (1 - out_{o1}) * out_{h1}$$

$$= (0.01 - 0.7459) * (0.7459)(1 - 0.7459) * (0.5932)$$

$$= +0.082$$ since value is -ve, we use (−) minus in formula.

$$w_{5new} = 0.4 - (0.5)(0.082)$$

$$= 0.4 - 0.041$$

$$= 0.359$$

for w_6 : $\frac{\partial E_{Total}}{\partial w_6} = -(\text{target} - out_{o1}) * out_{o1} * (1 - out_{o1}) * out_{h2}$

$$= -(0.01 - 0.7459) * (0.7459) * (1 - 0.7459) * 0.569$$

$= 0.08266$

$w_{5new} = (0.45) - (0.5)(0.08266)$

$= 0.45 - 0.04133 = 0.4086$

for w_7:- $\frac{\partial E_{Total}}{\partial w_7} = +(target - out_{o_2}) \cdot out_{o_2} * (1 - out_{o_2}) * out_{h_1}$

$= (0.99 - 0.7729) \cdot (0.7729)(1 - 0.7729) \cdot (0.5932)$

$= 0.0226$

$w_{7new} = w_7 - (0.5)\frac{\partial E_{Total}}{\partial w_7} = (0.5) - (0.5) \cdot (0.0226)$

$= 0.488$

for w_8:- $\frac{\partial E_{Total}}{\partial w_8} = +(target - out_{o_2}) * (out_{o_2})(1 - out_{o_2}) \cdot out_{h_2}$

$= (0.99 - 0.7729)(0.7729)(1 - 0.7729) \cdot (0.5928)$

$= 0.02259$

$w_{8new} = w_8 - \eta \cdot \frac{\partial E_{Total}}{\partial w_8}$

$= 0.55 - (0.5) \cdot (0.02259)$

$= 0.538705$

Hidden layers:-

$\frac{\partial E_{Total}}{\partial w_1} = \frac{\partial E_{Total}}{\partial out_{h_1}} * \frac{\partial out_{h_1}}{\partial net_{h_1}} * \frac{\partial net_{h_1}}{\partial w_1}$

$\frac{\partial E_{Total}}{\partial w_1} = -(target - out_{h_1}) * out_{h_1} \cdot (1 - out_{h_1}) * i_1$

$= (0.01 - 0.5932)(0.5932)(1 - 0.5932) * (0.05)$

$= 0.0070306$

$w_{1new} = w_1 - \eta \cdot \frac{\partial E_{Total}}{\partial w_1} = (0.15) - (0.5)(0.0070306)$

$= 0.146482$

$\frac{\partial E_{Total}}{\partial w_2} = (target - out_{h_1}) * (out_{h_1})(1 - out_{h_1}) * i_2$

$= (0.01 - 0.5932)(0.5932)(1 - 0.5932) * (0.10)$

$= 0.01407$

$w_{2new} = 0.2 - (0.5)(0.01407) = 0.1929$

for w_3: $\frac{\partial E_{Total}}{\partial w_3} = (target - out_{h_2})\, out_{h_2}(1 - out_{h_2})\, i_1$

$= (0.99 - 0.5969)(0.5969)(1 - 0.5969)(0.05)$

$= 0.004738$

$w_3 new = w_3 - \eta \cdot \frac{\partial E_{Total}}{\partial w_3}$

$= (0.25) - (0.5)(0.00473)$

$= 0.24763$

for w_4:-

$\frac{\partial E_{Total}}{\partial w_4} = (target - out_{h_2})\, out_{h_2}(1 - out_{h_2}) \cdot i_2$

$= (0.99 - 0.5969)(0.5969)(1 - 0.5969)(0.10)$

$= 0.009458$

$w_4 new = w_4 - \eta \cdot \frac{\partial E_{Total}}{\partial w_4}$

$= (0.3) - (0.5)(0.009458)$

$= 0.2952$

new $w_1 = 0.14$ new $w_5 = 0.35$

new $w_2 = 0.19$ new $w_6 = 0.40$

new $w_3 = 0.24$ new $w_7 = 0.48$

new $w_4 = 0.29$ new $w_8 = 0.53$

KNN (K-Nearest Neighbour)

3. Distances:- → Euclidean, → Manhattan, → Minkowski — dissimilarity

→ Cosine, → Jaccard — Similarity ⇒ how similar 2 vectors are.

$d(x_i, y_i) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$ ⇒ Euclidian distance

* shld be odd

This model calculation is complex for large dataset

It is lazy classifier — doesn't build model during training. It postpones learning until prediction is required

Suppose, 10,000 training samples, 20 features per sample. For 1 prediction, it performs thousands of distance calc involving 20 features.

Def ~~Minkowski~~ Manhattan distance :- $\sum_{i=1}^{n} |x_i - y_i|$

Minkowski distance :-

$$D(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^q\right)^{1/q}.$$

Cosine :- $\frac{A \cdot B}{\|A\| \|B\|}$ or $\frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2 \cdot \sum_{i=1}^{n} B_i^2}}$

Jaccard :- $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$

Jaccard distance = $1 - J(A, B)$

A (x_1, y_1) $(0, 4)$
B (x_2, y_2) $(6, 2)$
C (x_3, y_3) $(9, 1)$

Calculate distance using ~~Manhattan~~ Euclidean

$|0-4| + |6-2| + |9-1|$

$= 4 + 4 + 8 = 16$

Dist b/w A & B :- $\sqrt{(0-6)^2 + (4-2)^2} = \sqrt{36+4} = \sqrt{40} = 6.32$

" " A and C = $\sqrt{(0-9)^2 + (4-1)^2} = \sqrt{81+9} = \sqrt{90} = 9.486$

" " B and C = $\sqrt{(6-9)^2 + (2-1)^2} = \sqrt{9+1} = 3.162$

B & C are similar.

Manhattan:-

A & B = :- $|0-6| + |4-2| = 8$

B & C = $|6-9| + |2-1| = 4$

A & C = $|0-9| + |4-1| = 11$

B & C are similar

25/6/26

BMI	Age	Sugar	[label]
33.6	50	1	
26.6	30	0	
23.4	40	0	
43.1	67	0	
35.3	23	1	
35.9	67	1	
36.7	45	1	
25.7	46	0	
28.3	29	0	
31	56	1	

k = 3. Using KNN algorithm Predict class label for the following test record.

BMI	Age	Sugar
27.3	45	?

Distance matrix using Euclidean distance

$$d = \sqrt{\sum_{i=1} (x_i - y_i)^2}$$

Manhattan:-

$$d = \sum |x_i - y_i|$$

Applying euclidean:-

$\sqrt{(33.6-27.3)^2 + (50-45)^2} = \sqrt{64.69} = 8.04$

$\sqrt{(26.6-27.3)^2 + (30-45)^2} = \sqrt{225.49} = 15.01$

$\sqrt{(23.4-27.3)^2 + (40-45)^2} = \sqrt{40.21} = 6.34$

$\sqrt{(43.1-27.3)^2 + (67-45)^2} = \sqrt{249.64 + 484} = \sqrt{733.64} = 27.085$

$\sqrt{(35.3-27.3)^2+(23-45)^2} = 23.4094$

$\sqrt{(35.9-27.3)^2+(67-45)^2} = 23.6212$

$\sqrt{(36.7-27.3)^2+(45-45)^2} = 9.4$

$\sqrt{(25.7-27.3)^2+(46-45)^2} = 1.8868$ → 1st smallest

$\sqrt{(23.3-27.3)^2+(29-45)^2} = 16.4924$

$\sqrt{(31-27.3)^2+(56-45)^2} = 11.6056$

Manhattan :-

$d_1 = |33.6-27.3| + |50-45| = 11.3$ ~~2nd smallest~~

$= |26.6-27.3| + |30-45| = 15.7$

$= |23.4-27.3| + |40-45| = 8.9$ → 2nd smallest

$|43.1-27.3| + |67-45| = 37.8$

$|35.3-27.3| + |23-45| = 30$

$|35.9-27.3| + |67-45| = 30.6$

$|36.7-27.3| + |45-45| = 9.4$ → 3rd smallest

$|25.7-27.3| + |46-45| = 2.6$ → 1st smallest

$|23.3-27.3| + |29-45| = 20$

$|31-27.3| + |56-45| = 14.7$

for $k=3$, nearest neighbours are 8.04, 6.34, 1.8868

with label 1, 0, 0.

Itemset id ⇒

id	value	label
1	8.04	1
3	6.34	0
8	1.8868	0

∴ $k=0$ = class

Using manhattan :—

id	class label	value
3	0	~~2~~ 8.9
7	1	~~8~~ 9.4
8	0	2.6

∴ ~~K~~ = 0 ~~#~~

Confusion matrix

		Actual	
		y_1	y_2
Predicted	y_1	TP	FP
	y_2	FN	TN

Accuracy :- $\frac{TP + TN}{TP + TN + FP + FN}$

Precision :- (Out of all tves, how many are actually tve) $\frac{TP}{TP + FP}$

Recall :- $\frac{TP}{TP + FN}$

F1 Score :- $\frac{2 \cdot P \cdot R}{P + R}$

01/7/26

Unit-4 - Clustering Analysis

→ Basic Concepts
→ Partitioning methods
→ Hierarchical methods
→ Density Based methods
→ Grid-based methods
→ evaluation of clustering

cluster - grp of data objects, It is unsupervised learning

diff clustering approaches:-

Partitioning approach
Hierarchical approach
Density-Based "
Grid-based "
Model-based "
Freq-pattern-based "
User-guided or constraint-based "
Link-based Clustering "

Partitioning → Partition into set of k clusters.

$$E = \sum_{i=1}^{k} \sum_{p \in C_i} (p - c_i)^2$$

Partitioning criteria :-
→ Global optimal
→ Heuristic methods - k means & k mediods alg.

k means → Each cluster is represented by center of the cluster

k-Means clustering method → variation of k-means method.

Prob with K-Means Method.

k-medoid clustering

PAM (Partitioning Around Medoid)

efficiency improvement on PAM :-
→ CLARA
→ CLARANS

Datapts:-

1. (2, 10)
2. (2, 6)
3. (11, 11)
4. (6, 9)
5. (6, 4)
6. (1, 2)
7. (5, 10)
8. (4, 12)
9. (10, 12)
10. (7, 5)
11. (9, 11)
12. (4, 6)
13. (3, 10)
14. (3, 8)
15. (6, 11)

Cluster1 centroid (2, 6)
Cluster2 centroid (5, 10)
Cluster3 centroid (6, 11)

Three clusters to be formed using K means where K = 3

Using Euclidean distance:- (with 1st value)

for C1:- $\sqrt{(2-2)^2+(10-6)^2} = \sqrt{0+4^2} = \sqrt{16} = 4$

for C2:- $\sqrt{(2-5)^2+(10-10)^2} = \sqrt{9+0} = 3$ — min — this is centroid

for C3 = $\sqrt{(2-6)^2+(10-11)^2} = \sqrt{4^2+1^2} = \sqrt{17} = 4.123$

for 2nd value for C1:- $\sqrt{(2-2)^2+(6-6)^2} = 0$ — min — this is centroid

C2: $\sqrt{(2-5)^2+(6-10)^2} = 5$

C3: $\sqrt{(2-6)^2+(6-11)^2} = 6.40$

3rd value

C1: $\sqrt{(11-2)^2+(11-6)^2} = 10.29$

C2: $\sqrt{(11-5)^2+(11-10)^2} = 6.08$

C3 = $\sqrt{(11-6)^2+(11-11^2)} = 5$ — this is centroid

4th value:-

$C_1 = \sqrt{(6-2)^2+(9-6)^2} = 5$

$C_2 = \sqrt{(6-5)^2+(9-10)^2} = 1.414$ — this is centroid

$C_3 = \sqrt{(6-6)^2+(9-11)^2} = 2$

Illarly Calculate for all 15 values.

Datapt	C_1	C_2	C_3	Assigned cluster
(2,10)	4	3	4.123	C_2
(2,6)	0	5	6.40	C_1
(11,11)	10.29	6.08	5	C_3
(6,9)	5	1.414	2	C_2
(6,4)	4.47	6.08	7	C_1
(1,2)	4.123	8.944	10.29	C_1
(5,10)	5	0	1.414	C_2
(4,12)	3.6	1.44	2.8	C_2
(10,12)	10	5.38	~~4.123~~ 6.08	C_3
(7,5)	5.09	~~5.38~~ ~~4.123~~	6.08	C_1
(9,11)	8.60	4.123	3	C_3
(4,6)	2	4.123	5.385	C_1
(3,10)	4.123	2	3.1622	C_2
(3,8)	2.236	2.82	4.24	C_2
(6,11)	6.403	1.414	~~1.414~~ 0	C_3

$C_1 = \{2, 5, 6, 10, 12, 14\}$

$C_2 = \{1, 4, 7, 8, 13\}$

$C_3 = \{3, 9, 11, 15\}$

$C_1 \rightarrow$ mean :-

(2,6)
(6,4)
(1,2)
(7,5)
(4,6)
(3,8)

$\frac{23}{6}, \frac{31}{6}$

3.833, 5.167.

$C_2 \rightarrow$ mean :-

(2,10)
(6,9)
(5,10)
(4,12)
(3,10)

4 | 9.6

$C_3 \rightarrow$ mean

(11,11)
(10,12)
(9,11)
(6,11)

9, 11.25

∴ New Centroids :-

$C_1 = (3.833, 5.167)$, $C_2 = (4, 9.6)$, $C_3 = (9, 11.25)$

with new centroids :- for (2,10) :-

$c_1 :- \sqrt{(2-3.833)^2 + (10-5.167)^2} = \sqrt{3.359 + 23.425} = 5.17$

$c_2 :- \sqrt{(2-4)^2 + (10-9.6)^2} = \sqrt{4+0.16} = 2.03$

$c_3 = \sqrt{(2-9)^2 + (10-11.25)^2} = \sqrt{49+1.5625} = 7.11$

for (2,6) ⇒

$c_1 : \sqrt{(2-3.833)^2 + (6-5.167)^2} = 2.013$ — c_1

$c_2 : \sqrt{(2-4)^2 + (6-9.6)^2} = 4.118$

$c_3 = \sqrt{(2-9)^2 + (6-11.25)^2} = 8.750$

for (11,11) ⇒

$c_1 : \sqrt{(11-3.833)^2 + (11-5.167)^2} = 9.241$

$c_2 : \sqrt{(11-4)^2 + (11-9.6)^2} = 7.139$

$c_3 : \sqrt{(11-9)^2 + (11-11.25)^2} = 2.016$ — c_3

for (6,9) ⇒

$c_1 : \sqrt{(6-3.833)^2 + (9-5.167)^2} = 4.403$

$c_2 : \sqrt{(6-4)^2 + (9-9.6)^2} = 2.088$ — c_2

$c_3 : \sqrt{(6-9)^2 + (9-11.25)^2} = 3.75$

for (6,4) ⇒

$c_1 : \sqrt{(6-3.833)^2 + (4-5.167)^2} = 2.461$ — c_2

$c_2 : \sqrt{(6-4)^2 + (4-9.6)^2} = 5.946$

$c_3 : \sqrt{(6-9)^2 + (4-11.25)^2} = 7.846$

Nearly Compute for others.

Pt	C_1	C_2	C_3	Nearest Clusters
(4,2)	4.282	8.161	12.299	C_1
(5,10)	5.0	1.077	4.070	C_2
(4,12)	6.835	2.433	5.060	C_2
(10,12)	9.260	6.297	1.250	C_3
(7,5)	3.218	5.424	6.946	C_1
(9,11)	7.830	5.123	0.250	C_3
(4,6)	0.850	3.6	7.318	C_1
(3,10)	4.908	1.077	6.046	C_2
(3,8)	2.953	1.887	6.708	C_2
(6,11)	6.223	2.441	3.010	C_2

Cluster C_1: (2,6), (6,4), (1,2), (7,5), (4,6)

Centroid 1 = (4, 4.6)

Cluster C_2:- (2,10) (6,9) (5,10), (4,12), (3,10), (3,8), (6,11)

Centroid 2 = (4.143, 10)

Cluster C_3:- (11,11), (10,12), (9,11)

Centroid 3 = (10, 11.333)

Iteration 3:-

Point	C_1	C_2	C_3	Cluster
(2,10)	5.769	2.147	8.050	C_2
(2,6)	2.441	4.165	9.465	C_1
(11,11)	9.457	6.954	1.054	C_3
(6,9)	4.833	2.098	4.269	C_2
(6,4)	2.088	6.331	8.272	C_1
(1,2)	4.031	8.641	13.019	C_1
(5,10)	5.477	0.857	5.044	C_2
(4,12)	7.4	2.006	6.037	C_2

(7, 5)	3.027	5.6713	6.688	C_1
(9, 11)	8.101	5.030	1.054	C_3
(4, 6)	1.400	4.145	8.042	C_1
(3, 10)	5.494	1.143	7.088	C_2
(3, 8)	4.081	2.426	7.431	C_2
(6, 11)	6.707	2.141	4.177	C_2

New Clusters

Cluster 1	Cluster 2	Cluster 3
(2, 6)	(2, 10)	(11, 11)
(6, 4)	(6, 9)	(10, 12)
(1, 2)	(5, 10)	(9, 11)
(7, 5)	(11, 12)	
(4, 6)	(3, 10)	
	(3, 8)	
	(6, 11)	

$C_1 = (4, 4.6)$

$C_2 = (4.143, 10)$

$C_3 = (10, 11.333)$

The centroids didnot change, so the k-means algorithm has converged. This is the final iteration & no further iterations are required.

Next Question in next book.